

Self-Detect–Refine: Iterative Training Label Refinement Method for Clinical Note Generation

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Abstract

In medical LLM adaptation, collecting large-scale real clinical notes is constrained by patient privacy and annotation cost, so training pipelines often rely on notes synthesized by commercial LLMs. Those synthetic notes are not error-free. We observe that after fine-tuning on such notes, a task judge rates the resulting model’s outputs as more complete without a consistent Accuracy gain, while a hallucination judge flags more statements that the source does not support. We therefore report Accuracy and Completeness separately from hallucination counts, split the latter into Content and Format-Meta hallucinations, and report both as rates normalized by the number of fields each output format requires, so that a gain in one cannot hide a loss in the other. We then introduce Self-Detect–Refine, which applies an external LLM to detect and revise hallucinations in the training labels themselves rather than in outputs at inference time. We evaluate it across three base models, two refiners, and four refinement depths, fine-tuning a fresh copy of each base model on each label snapshot. GPT-5.2-refined labels lower Content hallucinations in every configuration, both as a share of the required fields and after length normalization. The two stages load on different components: the Completeness gain belongs to raw-label fine-tuning, whereas refinement raises Accuracy without a consistent loss in Completeness. Format-Meta changes are mixed, a lower-cost refiner is less consistent, and every judgment here comes from LLM judges on a single synthetic benchmark without clinician validation.

1 Introduction

Collecting real clinical notes at scale is constrained by patient privacy and annotation cost, so medical NLP pipelines increasingly train on notes synthesized by large language models (LLMs) (Tang et al., 2023; Kumichev et al., 2024). Clinical-note generation requires a completed form to represent the

supplied patient information accurately and completely, without adding unsupported details. We study this task on a synthetic English benchmark whose completed-note labels are themselves LLM-generated. Those labels can contain hallucinations, raising two questions: how does fine-tuning on them affect generated notes, and can refining them improve the resulting models?

A single judge-assigned score cannot separate accuracy from information coverage or from unsupported additions. We therefore use two judges. A *task judge* scores *Accuracy*, how faithfully the completed form reflects the source, and *Completeness*, how much source-supported information it fills in. We report the two separately throughout, since a change in one can offset a change in the other. A *hallucination judge* separately returns one record per unsupported or unfaithful statement, which we group into *Content* and *Format-Meta* and count. These counts reflect the statements the judge flagged, not clinician-verified errors. The task scores alone do not determine the hallucination counts, so we report both.

Self-Detect–Refine uses one external LLM to both detect hallucinations in a synthetic training label and revise the flagged content under a minimal-edit instruction; the *self-* prefix denotes this shared model, and the task and hallucination judges are different models again. We repeat the procedure to obtain label snapshots at different refinement depths, then fine-tune a fresh copy of each base model on each snapshot. We compare these models with the original base and with unrefined-label fine-tuning. Figure 1 summarizes the workflow.

Contributions.

- **Separate quality measurements.** We report Accuracy and Completeness separately from each other and from eight hallucination categories in two groups. Hallucinations are reported as the fraction of required fields carrying at least one,

a denominator fixed by the case rather than by output length or field coverage.

- **Training-label refinement pipeline.** We apply repeated detection and minimal-edit revision to the training labels rather than to inference outputs.
- **Quantitative observation of contamination and its refinement.** Three base models $\times$ two refiners $\times$ four depths, with aggregate and paired note-level results. The two stages separate by component: raw-label fine-tuning moves Completeness, refinement moves Accuracy.

2 Related Work

LLM hallucination detection and self-correction. LLM hallucination has been characterized by the intrinsic/extrinsic distinction (Maynez et al., 2020), surveys (Ji et al., 2023), and atomic-fact metrics (Min et al., 2023), with medical-domain benchmarks and detectors in Med-HALT (Pal et al., 2023) and ReXTrust (Hardy et al., 2025). On the self-correction side, Self-Refine (Madaan et al., 2023) proposed a single-LLM self-feedback framework, while Huang et al. (2024) found limits to intrinsic self-correction on reasoning tasks without external feedback. LLM-as-a-judge evaluation (Zheng et al., 2023) is itself known to carry position, verbosity, and self-enhancement biases. Our work differs in two respects: (i) the refinement target is the *training labels*, not inference outputs, and (ii) the detect-refine stage and the hallucination judge use different LLMs, separating model identity while retaining shared-rubric bias.

Data-centric AI and SFT label quality. Under the data-centric paradigm (Zha et al., 2023), one approach to SFT data quality is *high-quality sample selection* — LIMA (Zhou et al., 2023) attained strong alignment with 1,000 high-quality examples, AlpaGasus (Chen et al., 2024) filters data using LLM quality scores, DEITA (Liu et al., 2024) selects data by complexity, quality, and diversity, and CLEAR (Chen and Mueller, 2024) uses confidence estimates to filter or correct low-quality instruction-tuning data. Gekhman et al. (2024) showed in controlled closed-book QA that learning *new factual knowledge* through fine-tuning can increase hallucinations; we instead target the contamination *already existing in the training labels* and revise it through refinement rather than selection. Our procedure retains all samples.

Clinical note generation and synthetic medical data. Clinical note generation from doctor-patient encounters has been benchmarked by ACI-Bench (Yim et al., 2023) and MTS-Dialog (Ben Abacha et al., 2023); we adopt the same task format on synthetic inputs. To mitigate the medical data shortage, Tang et al. (2023) reported improved named-entity recognition and relation extraction after training on ChatGPT-generated labeled text, MedSyn (Kumichev et al., 2024) improved ICD classification with synthetic notes augmented by a medical knowledge graph, and Mishra et al. (2024) reduced summarization hallucinations via synthetic GPT-3.5/4 edit feedback with DPO and SALT. We instead quantify hallucinations in the synthetic labels themselves and measure hallucinations in the fine-tuned models’ outputs (§4.2).

3 Method

Figure 1a summarizes the full experiment, from source-data preparation to label refinement, independent fine-tuning, and evaluation. Original labels form $\mathcal{D}_0$; refinement produces snapshots $\mathcal{D}_K$. Each raw-label or refined-label SFT run starts from the original base checkpoint: the iteration-3 model does not continue training from the iteration-1 model. The Base condition receives no SFT. All conditions are evaluated on the same 200 inputs, disjoint from the 500-example refinement corpus.

3.1 Hallucination categories: Content and Format-Meta

The hallucination judge assigns each detected hallucination to one of eight categories, which we aggregate into two groups, Content and Format-Meta (FM); the categories and their group membership are listed in Table 1. Content covers patient facts, reasoning, quantities, and conversational context; Format-Meta covers template structure, unsupported reuse of prompt language, and internal-prompt or generation-context exposure. Separating these counts prevents gains in one group from obscuring deterioration in the other. The grouping is an analysis convention, not a clinical-risk scale. FM categories can carry clinical risk as well: prompt overgeneralization, one of the FM categories, can introduce clinical content, and Structural or Metadata errors can affect interpretation or workflow. Factual, medical-reasoning, and over-clinical categories may overlap conceptually; their boundaries and assignment reliability have not been

(a) Experiment workflow

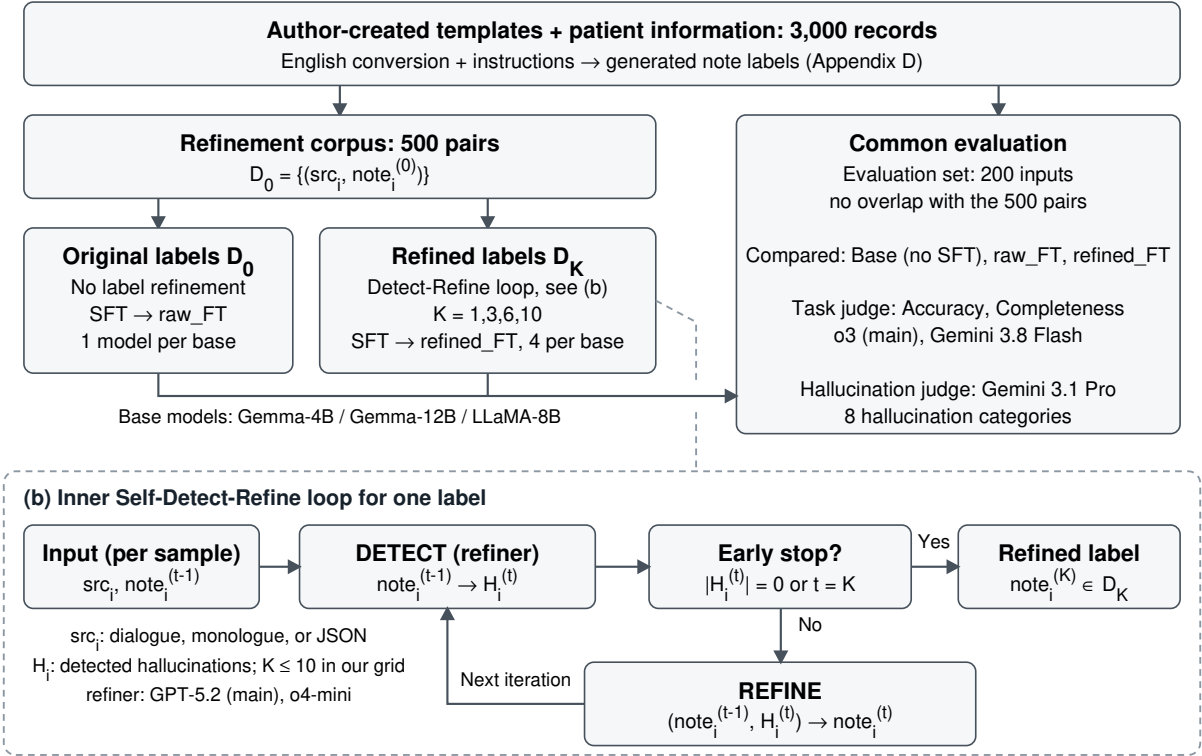


Figure 1: Experimental workflow and the inner refinement loop. **(a)** Data preparation, the refinement/evaluation split, snapshot construction, and the three compared model states. **(b)** The per-label detect-refine loop; early-stopped samples are carried forward, not dropped. Minimal edits and information preservation are prompt-level requests, not verified guarantees.

Cat	Name	Meaning
Content hallucinations (Content) — patient-related content		
F	Factual	Invented medical facts or patient attributes unsupported by the source
MR	Medical Reasoning	Unsupported clinical causality or judgment; treating tentative statements as definitive
OC	Overclinical	Inference from medical knowledge not stated in the source dialogue
N	Numerical	Distortion of times, numbers, ranges, or units
CC	Conv. Context	Misinterpreted dialogue context or incorrect attribution of speaker statements
Format-Meta hallucinations (FM) — template structure, prompt reuse, and generation-context exposure		
S	Structural	Content assigned to wrong fields; invented, duplicated, or deleted fields; deviations from the required structure
PO	Prompt Overgen.	Copying or generalizing prompt examples or phrasing without patient-specific support
ME	Metadata / Env.	Exposure or misuse of internal prompts, system information, or generation context

Table 1: Eight judge-assigned hallucination categories and their reporting groups, summarized from the detector prompt (App. H). Group membership is an operational convention, not a validated ranking of clinical severity; Content and Format-Meta errors can both have clinically relevant consequences.

validated by clinicians or a human agreement study.

3.2 Iterative Self-Detect-Refine pipeline

Starting from each synthetic note, the pipeline iteratively *detects* the remaining hallucinations and then requests *refinement* of the flagged content; iterating to a fixed depth K over the whole set yields a single refined training snapshot $\mathcal{D}_K$ per K , which is what we later fine-tune on (§4.1).

Inner loop (per sample, iteration t): (Fig. 1b)

- DETECT** — The LLM reviews the task prompt, the source input src_i , and the current note $note_i^{(t-1)}$, and extracts the hallucination set $\mathcal{H}_i^{(t)}$ (8-cat + violated_reason) via a required structured tool call. The prompt requests the hallucination type and a verbatim sentence or field as evidence with an explanation.

2. **REFINE** — Given $(\text{src}_i, \text{note}_i^{(t-1)}, \mathcal{H}_i^{(t)})$, the *same* LLM is instructed to revise or remove the listed errors using minimal edits, yielding $\text{note}_i^{(t)}$. Using one LLM for both detect and refine is what the *self*- prefix denotes.
3. **EARLY STOP** — Terminate the loop when the refiner’s DETECT returns $|\mathcal{H}_i^{(t)}|=0$ or on reaching the target depth ($K \leq 10$ in our grid). An early-stopped sample is not dropped: its last note is carried forward unchanged into every later snapshot, so K is the maximum depth a snapshot represents, not the number of edits applied to every sample.

3.3 Minimal-edit constraint of the refinement prompt

The refinement prompt asks the LLM to preserve correct, problem-free content and revise or remove only content identified in the error list. It requests minimal corrections, preservation of the original style, format, and structure as much as possible, and output of the revised note alone. The shared task prompt identifies patient information as the only factual source, user instructions as hard constraints, and the template as a fixed structure. These are prompt-level constraints, not mechanically enforced edit boundaries or guarantees of information preservation. The reference is the supplied synthetic input, not clinician-verified ground truth. Prompt details are given in App. H.

3.4 Downstream evaluation

We SFT each base LLM on every refined snapshot $\mathcal{D}_K$, producing one fine-tuned model per K , which we denote iter_K . Each fine-tuned model performs inference on the evaluation set, and primary evaluation uses the hallucination judge, which extracts 8-cat hallucinations via a structured tool call, and the task judge (Zheng et al., 2023). Both use models that differ from the inner-loop detect-refine LLM (specific model names in §4.1); the detection node and the Gemini hallucination judge use exactly the same task prompt, eight-category definitions, and required tool interface. This separates model identity while retaining a common rubric; shared-rubric bias and judge reliability are discussed in Limitations. The maximum refinement depth K controls the training snapshot; the selected iter_K snapshots form an evaluation grid for studying depth-dependent effects (see §4.1).

Item	Value
Training data	Author-written templates + patient info, translated to English with generated labels; 500 of 3,000 pairs (App. D)
SFT / internal val.	$n=450$ / $n=50$
Evaluation	$n=200$
Base LLMs	Gemma-4B, Gemma-12B (Gemma Team, 2025), LLaMA-8B (Grattafiori et al., 2024)
Refiner	GPT-5.2 (main), o4-mini (App. B); call settings in App. H
Hallucination judge	Gemini 3.1 Pro
Task judge	o3 (main), Gemini 3.8 Flash (App. M)
iter_K	$\{1, 3, 6, 10\}$
Hyperparams	QLoRA (Detrmers et al., 2023; Hu et al., 2022) SFT; configuration in Appendix G

Table 2: Experimental setup. Each snapshot is $\mathcal{D}_K = \{(\text{src}_i, \text{note}_i^{(K)})\}_{i \in \text{train}}$ (§3).

4 Experiments

4.1 Setup

Data preparation. We construct English input-label pairs from an author-created Korean corpus of scenario-based patient information and medical templates. Templates are generated using LLMs with public medical forms as references or derived from dialogues. We sample 3,000 form-based and dialogue records, translate templates and patient information into English, simplify template descriptions, and generate user instructions. English completed-note labels are newly generated by o4-mini, Claude Haiku-4.5, and Gemini-2.5-Flash-Lite. Inputs include dialogues, clinician or patient monologues, and JSON. Reproducibility details are in Appendix D.

Data partitions. We randomly sample 500 examples for label refinement and fine-tuning, and 200 evaluation examples with non-overlapping sample IDs from the 3,000-example pool (seed 123). The 500-example refinement corpus is split into 450 examples for parameter updates and 50 for internal validation used in early stopping and checkpoint selection. Label-contamination statistics and refinement costs refer to the full 500-example corpus. The 450/50 split and the model inputs are fixed across all conditions; only the completed-note labels change with refinement (App. G). Task scores and hallucination counts are reported on the 200-example evaluation set. Throughout the results, “evaluation set” refers to this 200-example set.

Evaluation grid. Table 2 summarizes the setup. For each $K \in \{1, 3, 6, 10\}$, the base LLM is

fine-tuned *separately* on the iteration-end training data $\mathcal{D}_K$, producing 12 parallel models per refiner ($3 \text{ base} \times 4 K$). The main text covers GPT-5.2 results (§4.3, §4.4); the o4-mini comparison is summarized in §4.5, with full details in App. B. Throughout, we compare three model states: **Base** (the un-fine-tuned base LLM), **raw_FT** (the base SFT-trained on the un-refined synthetic labels, i.e. snapshot $\mathcal{D}_0$), and **refined_FT** (the base SFT-trained on a refined snapshot $\mathcal{D}_K$, $K \geq 1$). Among the twelve GPT-5.2 refined_FT settings, we refer to Gemma-12B $\times$ iter₃, the cell with the lowest Content-rate point estimate on this evaluation set, as the *selected setting*; it is a reported operating point, not a validated optimum (§5).

Training configuration. All conditions use QLoRA (Detrmers et al., 2023) with 4-bit NF4 and training seed 42, on one RTX PRO 6000 GPU with Axolotl. Within each base model, raw_FT and every refined_FT condition use identical training settings, so comparisons between them are not confounded by hyperparameter differences; adapter and optimizer settings are tuned per base model (Appendix G). The seven-epoch limit is an upper bound; early stopping and checkpoint selection use internal validation loss.

Task rubric. The o3 task judge scores Accuracy and Completeness from 0 to 10 (higher is better). Results are reported per component; their mean is labelled Task and appears only in tables that likewise aggregate hallucinations into groups. A +0.63 Accuracy gain is a 0.63-point difference on this scale, without an established clinical-importance threshold. The full rubric is in Appendix F.

Paired bootstrap 95% CI. All Δ s (refined_FT – raw_FT, raw_FT – Base) carry a paired bootstrap 95% CI. We resample evaluation samples with replacement ($n=200$, $n_{\text{boot}}=1,000$; 10,000 where noted), keeping the two conditions paired within each resample. Records are summed without deduplication, not unique facts; Accuracy and Completeness are means; CIs are taken at the [2.5%, 97.5%] percentile. We mark significance (*) when the 95% CI excludes 0. Intervals are pointwise and unadjusted.

Field-normalized hallucination rate. Evaluation-side hallucination records are not reported as raw counts. We normalize them into a *hallucinated-field rate*: of the fields an output format requires, what fraction carries a hallucination.

The denominator is the number of fill-in fields in the case’s [FORMAT_TEMPLATE] segment, parsed deterministically and then manually verified for all 200 evaluation cases; it is a property of the case, so it is identical across Base, raw_FT and every refined_FT condition. The numerator is a per-field *presence* count, not a record count: each record is attributed to a field by its name or by locating its quoted span in the generated note, and a field is then counted *once* if at least one record of the hallucination type in question lands on it, however many records it carries. A field hit by both a Content and a Format-Meta record therefore counts once in each group, and the numerator can never exceed the denominator. Rates aggregate as $\sum \text{hallucinated fields} / \sum \text{required fields}$ and are reported as a percentage of the required fields, so differences between conditions are in percentage points (pp). A rate of 12.9% means that about 13 of every 100 fields the format requires carry a hallucination, and a difference of +5.5 pp means about 6 more fields out of every 100 do. Intervals are case-level paired bootstrap CIs computed as above; the train-side label statistics of §4.3 remain record counts.

4.2 Problem: contaminated labels and misleading gains

Train side: contamination of synthetic training labels. Content accounts for most of the hallucinations assigned by the hallucination judge (Gemini 3.1 Pro; 1,048 of 1,566 records; Fig. 12). The five Content categories are Overclinical (OC, 361), Factual (F, 301), Numerical (N, 225), Conv. Context (CC, 85), and Medical Reasoning (MR, 76); together they already appear in the synthetic labels at 2.1 per label on average. Format-Meta hallucinations are concentrated in Structural ($\Sigma=518$ records, S alone 421). In short, synthetic labels are not clean, and the Content categories dominate.

Evaluation side: misleading improvement relative to the base model. Under raw_FT, notes become more complete but also carry more judge-assigned Content hallucinations (Fig. 2). Completeness increases on all 3 bases (+0.51 to +1.25, all *), whereas Accuracy improves significantly only on LLaMA-8B (+0.74* [+0.53, +0.99]) and is inconclusive on both Gemma models (+0.23 [−0.10, +0.50] on Gemma-4B and −0.06 [−0.32, +0.22] on Gemma-12B); the additional Gemini 3.8 Flash task judge agrees in sign

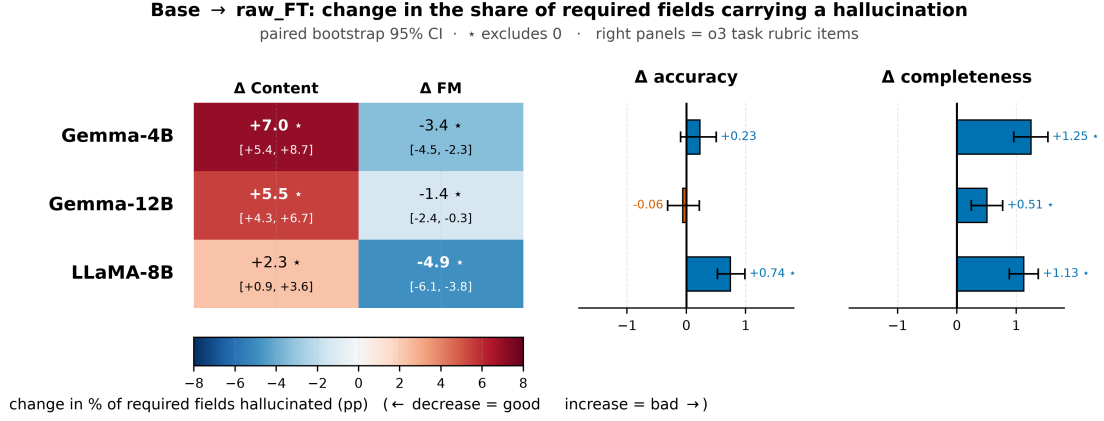


Figure 2: Paired-bootstrap 95% CI heatmap of (Δ Content, Δ FM) for raw_FT – Base; the two right-side panels share an x -scale. Cell = Δ [low, high]; * = CI excludes 0. Cells are the change in the share of required fields that carry a hallucination, in percentage points (§4.1) — a denominator fixed by the format. Gemma-12B reads +5.5* on Content: the share of its required fields carrying one rises 7.4% → 12.9%, about 6 more fields out of every 100 the format asks for. Remaining values and the task-side split are in §4.2, the 8-cat decomposition in App. A.

on all six component differences (Table 9). Content hallucinations increase on all 3 bases by +2.3 to +7.0 percentage points (pp) of the required fields (all *), with Gemma-12B going from 7.4% to 12.9% of its required fields — about 6 more fields out of every 100 the format asks for now carry a Content hallucination — while Format-Meta hallucinations decrease by 1.4 to 4.9 pp (all *).

The Content increase is not explained simply by longer or more complete outputs, because the reported quantity is already normalized: every rate below is the percentage of the fields the template requires that carry a hallucination (§4.1), a denominator that does not move between conditions, and Content hallucinations per 1,000 output characters also increase on all 3 bases (App. I).

Thus a single score combining Accuracy and Completeness would present raw_FT as an improvement on every base, largely driven by higher Completeness even where Accuracy does not improve and the Content rate increases; we call this a *misleading improvement*. This motivates refining the synthetic training labels before fine-tuning (§4.3), although the observed association does not establish a direct causal link between specific label and output hallucinations.

4.3 Refine effect

Train side: cleaning of the training labels. Two observations, using hallucination-judge counts at every round (Fig. 13). (i) Totals at iterations 0/1/3/6/10 are 1,566 → 1,017 → 880 → 835 → 804 — a fast-then-plateau pattern. (ii) Con-

tent hallucinations drop the most (1,048 → 592 → 461 → 419 → 391), whereas Format-Meta hallucinations move 518 → 425 → 419 → 416 → 413.

Evaluation side: fewer Content hallucinations at higher Accuracy. One refinement pass is already enough to move both sides. At iter₁ the hallucination-judge Content rate is lower than raw_FT on all 3 bases (−1.6 to −2.9 pp, all *), and the task-side gain sits in Accuracy: +0.525* [+0.332, +0.723] (Gemma-4B), +0.468* [+0.248, +0.668] (Gemma-12B), and +0.075 [−0.163, +0.325] (LLaMA-8B, inconclusive).

Completeness changes little and not consistently over the same three settings: +0.290* [+0.060, +0.555] (Gemma-4B), against +0.003 [−0.255, +0.258] (Gemma-12B) and +0.043 [−0.183, +0.285] (LLaMA-8B), both inconclusive. This is the mirror image of the raw_FT step (§4.2). Character-normalized counts point the same way, and output shortening alone does not explain the reductions (App. I).

Refinement depth: no single best K . Extending K -depth lowers the Content rate further relative to iter₁ on every base, along different trajectories (Fig. 3). Δ Content at iter_{1,3,6,10}, in pp, descends monotonically on Gemma-4B (−2.9 → −3.6 → −4.0 → −4.1) and LLaMA-8B (−1.8 → −2.4 → −2.5 → −3.0), whereas Gemma-12B (−1.6 → −5.0 → −4.7 → −4.5) is lowest at iter₃ and then rises slightly while staying well below iter₁.

Accuracy rises * in all 8 Gemma cells but

raw_FT → refined_FT: change in the share of required fields carrying a hallucination
paired bootstrap 95% CI · (gpt-5.2_iterK_FT, K ∈ {1,3,6,10}; n=200 × n_boot=1000) · * = CI excludes 0

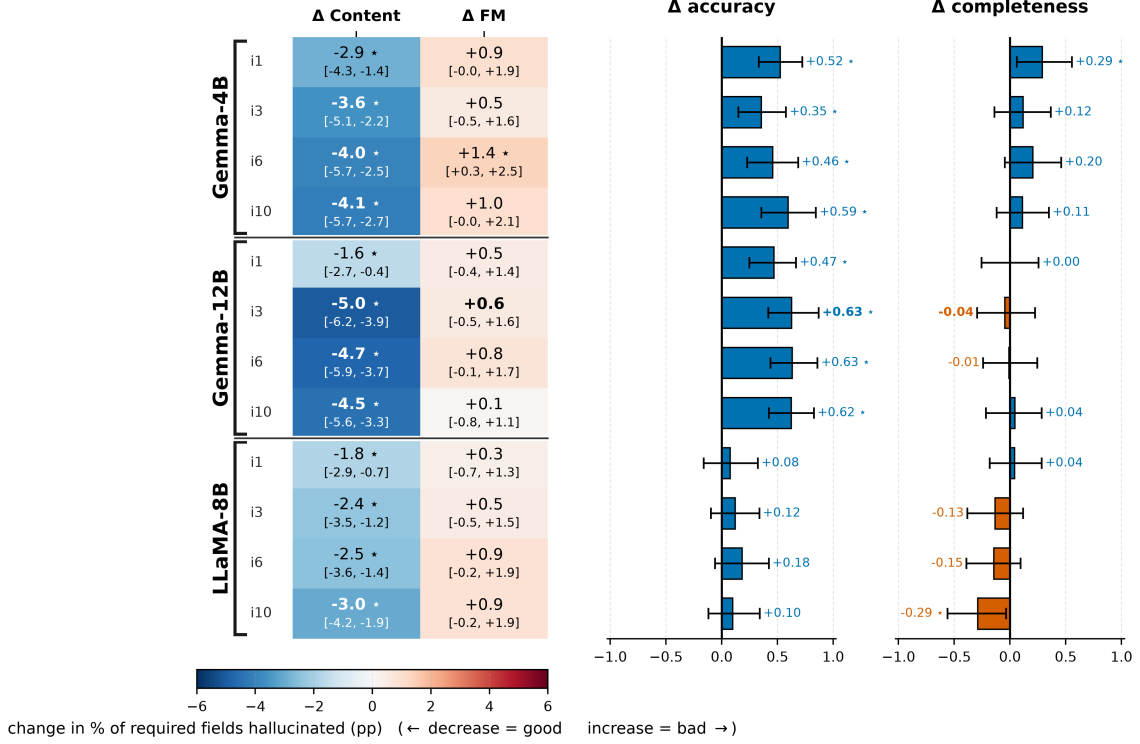


Figure 3: Paired-bootstrap 95% CI heatmap (columns: Δ Content, Δ FM) over the 12-cell grid (3 base $\times$ $K \in \{1, 3, 6, 10\}$), with the o3 task judge’s rubric items as right-side panels on a shared scale. Each cell = Δ with [low, high] below; * = CI excludes 0. Cells are the change in the share of required fields that carry a hallucination, in percentage points (§4.1). **Selected setting: Gemma-12B $\times$ GPT-5.2 iter₃** reads -5.0^* on Content (§4.3): 5 fewer fields out of every 100 carry one, which removes most of the raw_FT increase in Fig. 2. Per-cell outcomes are given in §4.3, the 8-cat decomposition in App. A (Fig. 4).

in none of the 4 LLaMA-8B cells, so the task-side recovery is base-dependent (Fig. 3). The grid also does not identify an optimal depth, because the three quantities peak at different depths. On Gemma-12B the Content rate is lowest at iter₃, whereas o3 Accuracy means at iterations 1/3/6/10 are 8.103/8.263/8.268/8.260 (highest at iter₆) and Completeness means are 7.603/7.555/7.590/7.645 (highest at iter₁₀). These rankings do not establish significant differences between iteration depths, and the depth that maximizes one quantity need not maximize another.

Train-side residuals keep falling modestly at deep iterations while the downstream marginal gain on Gemma-12B flattens after iter₃, which motivates the cost comparison in §5.

Component-level analysis of the selected setting.

For the selected Gemma-12B iter₃ setting, refinement removes most of the Content increase that raw-label fine-tuning introduced: the hallucination-

judge Content rate falls from 12.9% to 7.9% of the required fields (-5.0 pp, 95% CI $[-6.2, -3.9]$) — 5 fewer fields out of every 100 carry a Content hallucination, approaching the Base rate (7.4) without passing it in point estimate, at a refined-minus-Base difference the paired interval does not separate from zero (§5). It does so with higher Accuracy rather than with less content: on the o3 task judge Accuracy rises $+0.628$ (95% CI $[0.405, 0.860]$, excluding zero), while the Completeness difference is inconclusive (-0.045 , $[-0.305, 0.230]$) at numerically similar means (7.600 versus 7.555), which establishes neither equivalence nor information preservation. The Format-Meta difference is $+0.6$ pp (95% CI $[-0.5, +1.6]$), allowing improvement or worsening. These additional paired intervals use 10,000 resamples and seed 20260907; Appendix Table 12 reports the Accuracy and Completeness means, Appendix F covers all GPT-5.2 conditions and the note-level paired outcomes, Appendix L the counts of notes with no detected Content hallu-

cination, and Appendix E an illustrative case.

4.4 Joint Content–Format–Meta hallucination landscape

Arranging the 12 cells of §4.3 on the plane of Content hallucination reduction against Format–Meta change, both from the hallucination judge (Fig. 14), places every cell in the Content-reduction half-plane with its horizontal interval excluding 0. Only Gemma-4B iter₆ pays for that reduction with a * Format–Meta increase; the other 11 Format–Meta intervals are inconclusive, including the selected Gemma-12B iter₃ cell at (−5.0, +0.6).

4.5 Refiner ablation: a lower-cost refiner

We contrast a higher-cost refiner (GPT-5.2, high reasoning) with a lower-cost alternative (o4-mini, medium) under common downstream training and evaluation settings. This practical comparison does not isolate model identity from reasoning configuration. Only 5 of the 12 cells have a * Δ Content, against 12/12 for GPT-5.2. On Gemma-12B, o4-mini has its lowest observed hallucination-judge Content rate at iter₆ (−3.1* pp), compared with −0.3 pp at iter₁₀. Its highest o3 Accuracy mean occurs at iter₆ (8.130) and its highest o3 Completeness mean at iter₃ (7.925), so no single depth is best on both; the additional Gemini task judge peaks at iter₆ on Accuracy and at iter₁₀ on Completeness (Appendix M). The observed train-side count reduction is smaller: by iteration 10 the total falls 1,566 → 1,258 and Content hallucinations 1,048 → 809, against 804 and 391 for GPT-5.2. Full details are in App. B (Fig. 6–9, Table 3).

5 Discussion

Selected setting and scope. The selected Gemma-12B/GPT-5.2 iter₃ setting (§4.1) was chosen on the same evaluation set on which it is reported, and the o3 rubric items peak at other depths (§4.3), so it remains a reported operating point rather than a validated optimum. Differences across the three bases likewise do not establish a capacity ordering, because model family, size, and tuned hyperparameters vary together.

Comparison with the un-fine-tuned base. Content alone understates the comparison against Base. On Gemma-12B the total field rate is 16.5 (Base), 20.3 (raw_FT), and 16.3 (GPT-5.2 iter₃): raw-label fine-tuning trades Format–Meta errors for

Content ones, and Base is both the least format-compliant of the three (FM 9.6% of required fields against 8.2% and 8.8%) and the lowest on both rubric items (Accuracy 7.70, Completeness 7.09, against 8.26 and 7.55 after refinement). The paired bootstrap of §4.1 leaves every refined-minus-Base Gemma-12B group inconclusive (Total −0.3 [−1.6, +1.0], Content +0.5 [−0.5, +1.5], FM −0.8 [−1.9, +0.2]), and the other bases diverge: LLaMA-8B iter₁₀ falls below its Base on the total rate (−4.9* [−6.2, −3.6]) with Content inconclusive, whereas Gemma-4B iter₁₀ keeps a Content excess (+2.9* [+1.5, +4.2]) at an unchanged total (+0.1 [−1.4, +1.6]). Base is still the lower-Content condition in point estimate on two bases, but reaches that at lower judge-assessed coverage and weaker template compliance, not by producing better-grounded notes. These pointwise intervals are unadjusted for the selection and multiplicity described in Limitations.

Why the two stages load on different components. The split follows from what each stage can change. Raw-label fine-tuning supplies the output convention (which fields exist and how they are filled); this is what the Completeness item scores, and it is also the channel through which the labels’ unsupported content enters. Refinement is constrained by prompt to minimal edits over an existing label (§3.3): it is asked to delete or correct unsupported statements, not to add source-supported content the label omitted. It therefore cannot be expected to raise coverage, only the share of retained statements the source supports, which is what Accuracy scores. The measurements follow that division: Completeness rises at the raw_FT step on all three bases and is inconclusive in ten of twelve refined cells, whereas Accuracy is inconclusive on two bases at that step and rises * in all eight Gemma refined cells. This reads the design rather than testing it; a refiner permitted to add omitted content, or a larger edit budget, need not show the same split.

Conservatism as an alternative explanation. Fewer judge-assigned hallucinations can also be produced by writing less. Three observations bound that explanation without eliminating it. First, the hallucination rates are normalized by the number of required fields, a denominator fixed by the case (§4.1) rather than by how much the model chose to write. Second, the reduction is distributed rather than concentrated in a few notes: at the se-

lected setting 114 of 200 notes carry fewer Content records against 25 with more (App. F), and the share of notes with no detected Content record rises from 14.5% to 27.0%, against 25.5% for Base (App. L). Third, Content hallucinations per 1,000 output characters fall in all 12 GPT-5.2 settings (on Gemma-4B even as mean length rises), whereas with o4-mini this density falls in six settings and rises in six (App. I). None of these settles whether particular source-supported facts were dropped. LLaMA-8B iter₁₀, the GPT-5.2 setting with the largest Content reduction on that base, is also the only GPT-5.2 setting with a * lower Completeness under o3.

Judge sensitivity and completeness. On identical outputs and prompts, the two task judges (o3 and Gemini 3.8 Flash) both show Accuracy gains in 22/24 refined conditions, whereas Completeness rises under both in only 13/24 and falls under both in 5/24. The two components part company this often, so a lower Content rate does not uniformly accompany lower Completeness. Under the additional judge the selected setting's Completeness difference is significantly positive (+0.508* [+0.247, +0.792]) where o3 leaves it inconclusive (−0.045), so the Accuracy gain is reproduced by both judges while the direction of the coverage change is judge-dependent; omissions and length effects remain possible (Appendices M and I).

Cost and practical interpretation. GPT-5.2 iteration 3 records \$83.83 for final successful DETECT and REFINE responses across 500 labels and leaves 880 train-side hallucination records, compared with 1,258 after o4-mini iteration 10 at \$80.56 (Table 10; supporting plots in Appendix K). Appendix C compares, for each base and refiner, the setting with the lowest Content point estimate in the evaluated grid; what the cost totals include is stated with Table 10, and the baselines the comparison lacks are listed in Limitations.

6 Conclusion

Fine-tuning on synthetic clinical-note labels raises judge-assessed Completeness on all three base models while significantly increasing the Content hallucination rate, so a single averaged task score would present raw-label fine-tuning as an improvement over every base. Iterative label refinement with GPT-5.2 lowers the Content rate significantly

in every tested configuration and, at the selected setting, reverses most of that increase, leaving the total hallucination rate at a value the paired interval does not separate from Base's, without beating Base on Content alone. The task-side change that accompanies it is a gain in Accuracy rather than in Completeness, whose direction is model- and judge-dependent, and Format-Meta changes are inconclusive in nearly every setting. The two stages therefore act on different components, and a single score would have averaged the difference away. Reporting the two rubric items separately and normalizing hallucination records by a denominator the case fixes rather than by the output is what makes these movements visible; whether they carry over to real clinical notes, other templates, or clinician judgment remains to be tested.

Limitations

Measurement and clinical validity. Hallucination counts come from the hallucination judge, and Accuracy and Completeness from the task judge; there is no clinician evaluation or human agreement study. An additional task judge assesses the same outputs (App. M), but hallucination counts are not re-evaluated by a second judge, and the two task judges agree on the sign of the per-note Completeness change in only 50.3% of paired notes. The taxonomy is not a validated clinical-risk measure, Medical Reasoning and Overclinical Reasoning were assigned without disambiguation rules, and the detector and the hallucination judge, though different models, share the same prompt, category definitions, and tool interface, so agreement with this common rubric neither validates the categories nor excludes shared systematic errors. We do not isolate whether a Format-Meta deviation originated in label refinement or in downstream generation, and the single qualitative example (App. E) rests on no multi-note annotation and cannot establish clinical benefit.

Completeness and causal interpretation. Neither the Completeness item nor the field denominator measures verified recall of source-supported facts: both score coverage against what the template or the judge asks for. The selected setting is itself 14.3% shorter than raw_FT in mean characters. Beyond the o3 Completeness criterion we report no required-field fill rate, no preservation rate for source-supported information, and no separation of appropriately empty fields from true omissions,

so more conservative completion or deleted content may still account for part of the reduction (§5). We do not trace specific label hallucinations into model outputs and therefore do not establish causal amplification; an independent completeness audit is still needed.

Field-normalized rate. The denominator, parsed from each template, is identical across Base, raw_FT, and every refined_FT condition. The numerator attributes each hallucination record to a field by its name or a quoted span and collapses records per field, so a change in how many hallucinations share one field registers as no change in the rate. Record counts remain reported alongside the rate, and the two scales should not be compared cell by cell.

Selection and stochastic variation. The 200-example evaluation set does double duty: refinement depths are compared on it and the selected setting was chosen on it, so the depth comparison is exploratory and the set is not an independent test set for the selected setting. The pointwise bootstrap intervals are unadjusted for that selection and multiple comparisons; they describe evaluation-sample variation conditional on fixed outputs, not variation across training seeds, refinement runs, or judge calls. The scale is small throughout — 450 examples for parameter updates, 50 for internal validation, 200 for evaluation — which leaves per-category comparisons underpowered, and the experiments use one training seed and one hyperparameter setting per base. The grid trains only $K \in \{1, 3, 6, 10\}$, so neither an optimal depth nor the location of a plateau is established and the isolated o4-mini Gemma-12B iter₆ improvement may be an artifact of that spacing; two refiners do not establish a general effect of refiner strength. An interval that crosses zero records a difference the experiment did not detect; it is not evidence of equivalence or of absence of harm.

Scope and missing comparisons. The corpus is one task in one language: English hospital forms translated from an author-created Korean collection. Generalization to Korean, to other clinical document types, or to non-medical domains is untested, as is the taxonomy’s reproducibility on another dataset. The split is disjoint by sample ID, but similar templates may occur across partitions, so generalization to unseen templates is not established, and translation fidelity was not inde-

pendently verified. Training and evaluation labels come from the same synthetic generators, so the measured effect is a reduction within those generators’ hallucination distribution; no real-data or out-of-distribution evaluation is reported. Direct label regeneration, filtering, deletion-only refinement, and compute-matched single-pass baselines are absent, so the incremental benefit of iteration is unresolved. The selected setting remains above Base in the Content rate in point estimate. Reported API costs exclude failed attempts and retries, and do not measure total execution costs or comparative clinical utility.

Ethics Statement

Data and privacy. The source template and patient-information collection was created by the authors before this study. Patient information is synthetic, generated from designed scenarios rather than taken from real patient records (App. D). This study translates the resulting synthetic patient information and templates into English and generates completed-note labels using LLMs (App. D). The benchmark evaluates synthetic clinical-document inputs. **Intended use and clinical safety.** Task-score gains can co-occur with increased Content hallucinations, making clinical safety assessment a separate requirement. The pipeline is a research tool for cleaning training labels, not a clinical decision-support system, and any real-world deployment would require clinician oversight, prospective validation, and appropriate regulatory review. **Evaluation scope.** Hallucination reduction is measured by the hallucination judge rather than by clinically validated outcomes (see Limitations); reported reductions should be interpreted as judge-defined relative changes under a fixed evaluator distinct from the refiner, not as guarantees of factual correctness. **Artifacts and licensing.** Base models are subject to their respective licenses. Release plans for code and supplementary artifacts have not been finalized.

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A Bootstrap CI methodology and eight-category decomposition

Bootstrap caveats. A significant refined-minus-raw comparison for one refiner and a non-significant comparison for another do not themselves establish a significant difference between refiners; that requires a direct contrast. All significance markers in the figures use pointwise intervals, not simultaneous coverage of the grid.

Field denominator. The field-normalized rate (§4.1) has 3,623 fields over the 200 evaluation cases, 18.1 per case, identical across all conditions.

Eight-category decomposition. The 12-cell $\times$ 8-cat decompositions of the per-cell Δ from §4.3 are Fig. 4 (GPT-5.2) and Fig. 5 (o4-mini), both in percentage points of the field rate defined in §4.1. Cell = Δ^* [low, high]; bold row = selected setting.

B Additional results for o4-mini

We report the corresponding analyses for the o4-mini refiner; each subsection below mirrors the corresponding GPT-5.2 panel.

B.1 Train-side cleaning

Figure 6 tracks how o4-mini cleans the training labels across iterations 0, . . . , 10.

B.2 Evaluation-side bootstrap CI heatmap

Figure 7 gives the paired-bootstrap 95% CI heatmap over the 12-cell o4-mini grid.

B.3 (Δ Content, Δ Format-Meta) quadrant

Figure 8 places the 12 o4-mini cells on the (Δ Content, Δ FM) plane.

B.4 12-cell summary

Table 3 summarizes the 12 o4-mini cells (Δ Content and Δ task vs. raw_FT); it is grouped with the per-base best- K figure and table (Fig. 10, Table 4).

B.5 Refiner head-to-head (Gemma-12B)

Figure 9 compares the two refiners’ settings selected by minimum Content against Base and raw_FT on Gemma-12B.

C Selected per-base comparisons

Figure 10 and Table 4 compare settings selected by the lowest Content point estimate within the evaluated grid for each base and refiner.

D Training-data schema

Data source. We use a private, author-created corpus of synthetic patient information and medical templates, built by the authors before this study, sampling 3,000 form-based and dialogue records with random seed 123. The dataset is not publicly released, which limits independent reproduction of the results. Patient information is generated from designed scenarios rather than taken from real patient records, with distinct combinations of attributes such as sex, severity, specialty, disease, and monologue/dialogue format producing sample-specific content; absence of duplicate patient information has not been independently audited. Templates are either form-structured or derived from dialogues. Splits enforce sample-ID non-overlap only, with no restriction on template identity or similarity.

English preparation and labels. The translation prompt converts TEMPLATE and PATIENT_INFORMATION together, requiring English-only text, consistent terminology, and preserved structure. A regular-expression step retains template descriptions, after which a template-conditioned prompt generates user-style instructions. The 3,000 inputs are randomly shuffled and partitioned into three disjoint groups of 1,000, one per label generator listed in the main text. These models newly complete the forms in English; Korean assistant notes are neither translated nor included in the final English dataset. Before the DETECT-REFINE experiments, the generated English notes were used as raw labels without content-quality filtering, regeneration, or manual correction. The fill-in prompt treats patient information as the sole factual source, preserves template structure and field names, replaces examples with case content, and follows instruction-specific missing-value rules or default placeholders. These are prompt requirements, not

raw_FT → refined_FT: change in the share of required fields carrying a hallucination
paired bootstrap 95% CI · (gpt-5.2_iterK_FT, K ∈ {1,3,6,10}; n=200 × n_boot=1000) ·
cell = Δ [low, high] · * = CI excludes 0

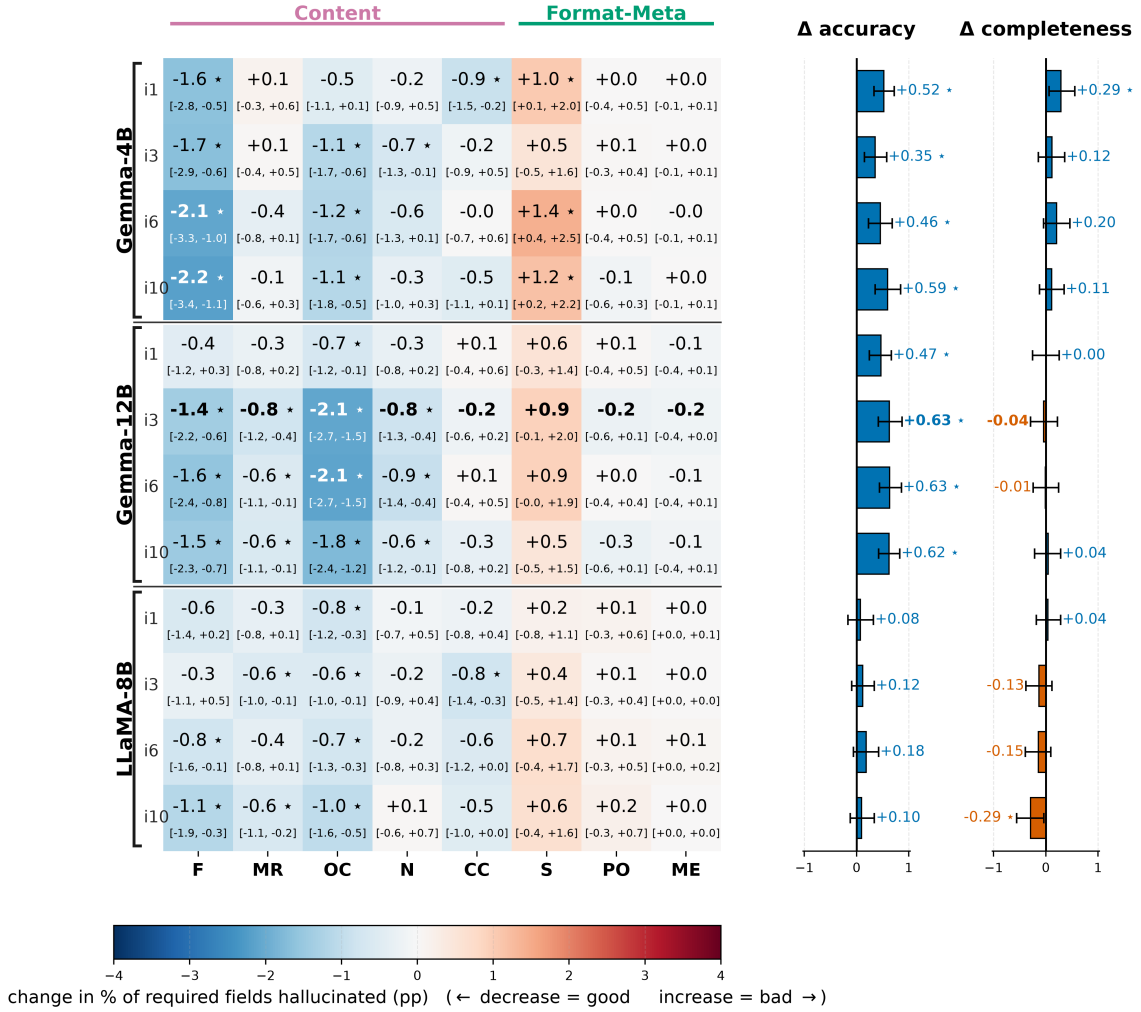


Figure 4: 8-cat decomposition heatmap of Fig. 3 (GPT-5.2, 2-group aggregate) in the main text, on the same field-rate scale, in percentage points (§4.1). Cell = Δ with the paired-bootstrap 95% CI [low, high] below; * = significant (CI excludes 0); the two right-side panels give the o3 task judge’s rubric items on a shared scale, matching Fig. 3. The selected setting (Gemma-12B × iter₃) Content reduction is jointly *across F / MR / OC / N (only CC is n.s.), and its task gain sits entirely in Accuracy (+0.63*) with Completeness inconclusive (−0.04). Gemma-4B shows S * in the *positive* direction at iter₁, iter₆, and iter₁₀ (+1.0*/ + 1.4*/ + 1.2* pp). At the group level the only *FM cell is Gemma-4B iter₆ (Fig. 14); category-level and group-level significance need not coincide, since offsetting PO and ME differences enter the FM total.

verified guarantees of correctness or information preservation. No separate post-translation quality review was performed.

Each sample consists of **3 separated input sections** and a **filled clinical note (label)**.

[INSTRUCTIONS] Hard constraints on clinical-note writing (format / tone / abbreviation rules / handling of missing information, etc.). Each sample is synthesized with a different set of rules.

[INPUT_SENTENCES] The *sole factual source* for

patient information. Mixed formats include clinician–patient dialogues, clinician or patient monologues, and JSON-formatted information. Hallucinations of F / MR / OC / N / CC (Content) from §3.1 are judged against this section.

[FORMAT_TEMPLATE] The fixed structure of the medical document to be filled (headers / tables / sections). Each field is annotated with a format-hint sentence; the model is expected to fill in patient data and then remove the hint itself. Structural er-

raw_FT → refined_FT: change in the share of required fields carrying a hallucination
paired bootstrap 95% CI · (o4-mini_iterK_FT, K ∈ {1,3,6,10}; n=200 × n_boot=1000) ·
cell = Δ [low, high] · * = CI excludes 0

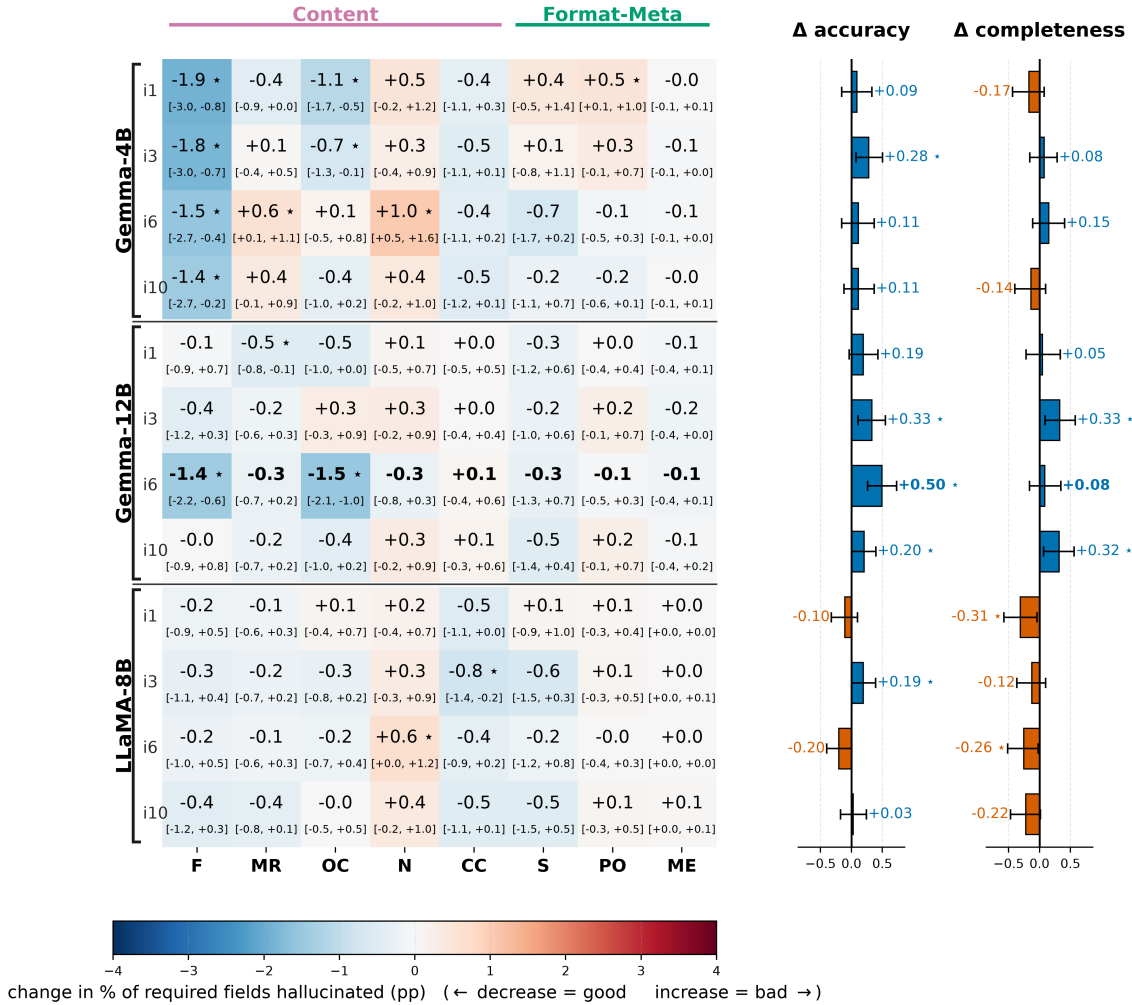


Figure 5: 8-cat decomposition of Fig. 7 (o4-mini), on the same field-rate scale, in percentage points (§4.1). Cell = Δ with the paired-bootstrap 95% CI [low, high] below; * = significant (CI excludes 0); the two right-side panels give the o3 task judge’s rubric items on a shared scale, matching Fig. 7. The Gemma-12B × iter₆ spike concentrates in F / OC at the cell level (−1.4* and −1.5* pp), so about 3 of the 100 required fields stop carrying a factual or over-clinical hallucination while the other six categories do not move. On the same base, iter₃ and iter₁₀ are 8-cat n.s. and iter₁ moves only MR (−0.5* pp) — evidence that the spike is cat-level single-shot.

rors are assessed against this template. Prompt Overgeneralization concerns unsupported reuse of prompt or example language; Metadata / Environment concerns internal-prompt, system, or generation-context exposure across the supplied context.

These input sections guide evaluation but do not partition errors one-to-one: the Content/FM groups in §3.1 are operational reporting groups, and category definitions may overlap.

FORMAT_TEMPLATE example (emergency assessment form).

```
I. Patient Info
1. Name : Sentence format: full legal name
2. Age/Sex : described as 62/F format
3. Visit Date & Time :
   YYYY-MM-DD HH:MM 24-hour format
II. Clinical Data
1. Chief Complaint : List with hyphens (-)
2. History of Present Illness : Sentence
3. Past Medical History :
   Separated by commas
...
```

E Qualitative case study

This case is an imaging report for an IPF patient (Gemma-12B + GPT-5.2 iter₃). Under raw_FT

Hallucination-judge records in training labels (iterations 0-10, n=500 labels)

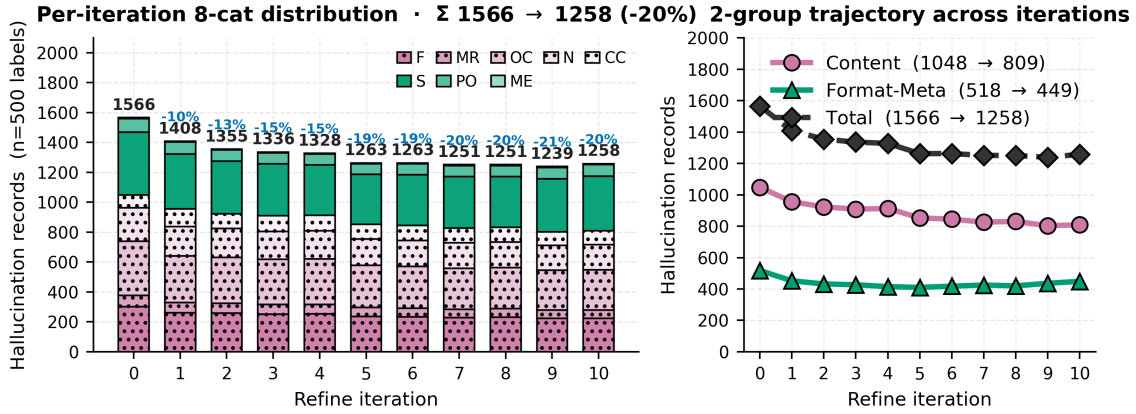


Figure 6: Hallucination-judge 8-cat evolution of training labels over o4-mini refinement iterations 0, ..., 10 (same layout as Fig. 13; Content group dot-hatched, Format-Meta solid). Totals at iterations 0/1/3/6/10 are Σ 1,566 $\rightarrow$ 1,408 $\rightarrow$ 1,336 $\rightarrow$ 1,263 $\rightarrow$ 1,258, against 804 for GPT-5.2 at iteration 10. Content hallucinations move 1,048 $\rightarrow$ 956 $\rightarrow$ 909 $\rightarrow$ 845 $\rightarrow$ 809, against 391 for GPT-5.2, and Format-Meta hallucinations 518 $\rightarrow$ 452 $\rightarrow$ 427 $\rightarrow$ 418 $\rightarrow$ 449 return toward their round-0 count by iteration 10. These are hallucination-count changes, not measurements of how much label text changed.

the model imports four non-imaging tests into the imaging-only template and fabricates the imaging-specific fields they lack, giving 5 Content hallucinations; the iter₃ model emits none of the 5, while Format-Meta hallucinations move 3 $\rightarrow$ 4. The case is illustrative and not statistically representative. Figure 11 shows the raw_FT and refined_FT outputs side by side.

Sample-level hallucinations and task scores.

Group	Base	raw_FT	iter ₃
Content (F/MR/OC/N/CC)	0	5	0
FM (S/PO/ME)	3	3	4
Σ	3	8	4
Accuracy (o3)	8	6	8
Completeness (o3)	6	3	3
Task (o3)	7.0	4.5	5.5

The stored o3 scores show that refinement improves Accuracy from 6 to 8 and Task from 4.5 to 5.5, while Completeness remains at 3, below Base’s 6. Thus fewer Content hallucinations do not establish a complete or clinically usable note. This example illustrates why task components and both hallucination groups should be considered together.

The 5 Content hallucinations raw_FT introduces and what the iter₃ output does instead.

(1) [OC] — adds the model’s own clinical interpretation in the X-ray Remarks (“cardiomegaly may be related to pulmonary hypertension”); the iter₃ output carries no such interpretation, leaving only

a comparison note in the Remarks. (2) [F] — in ABGA Remarks, “on room air” is a factual error (the input states “on 2 L oxygen”); the iter₃ output does not list the ABGA item. (3) [F] — interpreting physicians (“Lee Sang-min,” “Kim Young-su”) assigned to Inflammatory markers / ABGA / Biochemistry / PFT, which as lab/functional panels carry no interpreting physician in the input (the names are borrowed from the ordering physician and the CT reader); the iter₃ output does not list those examinations. (4) [F] — a fabricated PFT order date/time. (5) [N] — a Biochemistry examination-time error (15:00 instead of the specimen collection time, 14:50).

Observation. raw_FT’s 5 Content hallucinations arise from two paths — (a) filling X-ray / HRCT Remarks with its own clinical interpretation (1 OC) and (b) importing non-imaging tests that the input provides only as laboratory/functional results (Inflammatory markers / ABGA / Biochemistry / PFT) and reporting them as imaging examinations, packing fabricated interpreting physicians / date-times / an “on room air” status into them as factual errors. the iter₃ output avoids all 5 at once by *listing only the imaging examinations and leaving the X-ray Remarks uninterpreted*.

FM observations ($n=1$). The refined_FT output has 4 detected FM hallucinations: the three also detected for Base plus “Remarks: None” in-

raw_FT → refined_FT: change in the share of required fields carrying a hallucination
paired bootstrap 95% CI · (o4-mini_iterK_FT, K ∈ {1,3,6,10}; n=200 × n_boot=1000) · * = CI excludes 0

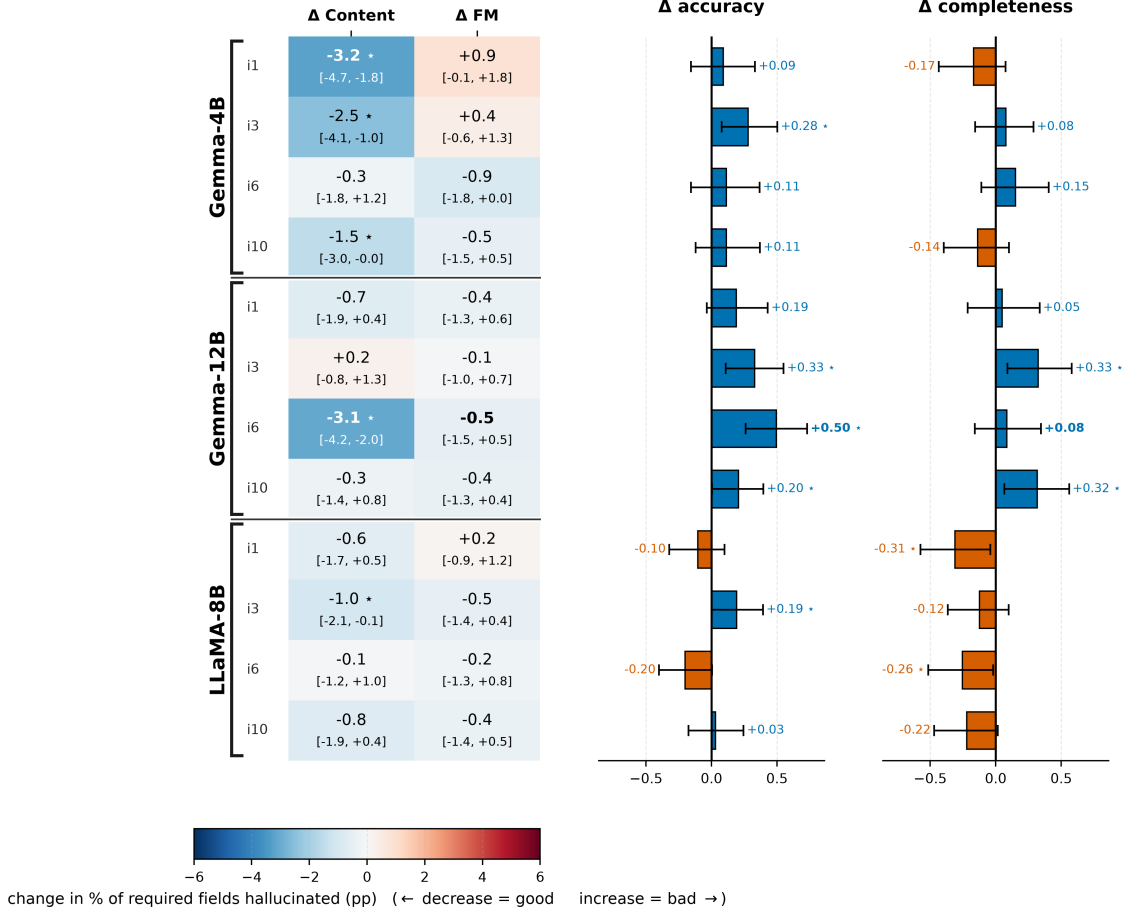


Figure 7: Paired-bootstrap 95% CI heatmap (2 columns: Δ Content, Δ FM) over the 12-cell grid (3 base $\times$ o4-mini $\times$ $K \in \{1, 3, 6, 10\}$). Cells are the change in the share of required fields that carry a hallucination, in percentage points (§4.1). Lowest observed Content rate = Gemma-12B $\times$ iter₆, Δ Content = $-3.1^* [-4.2, -2.0]$ pp ($-128^* [-174, -86]$ records) — 3 fewer fields out of every 100 carry one, about 60% of what GPT-5.2 removes at its selected setting — with an FM point difference of -0.5 (-23 records) whose CI includes zero. The same base’s Content CIs at iter₁, iter₃, and iter₁₀ include zero on both scales (-0.7 , $+0.2$, -0.3 pp). Normalization leaves the ablation’s pattern intact: the same 5/12 Content cells are * as on record counts, and FM is * in 0/12 rather than 1/12 (Gemma-4B iter₆, $-0.9 [-1.8, +0.0]$, is the borderline cell). The two right-side panels split the o3 task score into its rubric items on a shared scale (as in Fig. 3). Content is lowest at iter₆, where Δ Accuracy is largest ($+0.50^*$) but Δ Completeness is inconclusive ($+0.08$); at iter₃ both items rise by $+0.33^*$. Unlike GPT-5.2, neither item moves consistently across K : Accuracy is * in 5/12 cells and Completeness in 4/12, the latter split between two Gemma-12B increases and two LLaMA-8B decreases.

stead of the instructed “—”. These are downstream model outputs; this comparison does not isolate whether the deviations originated in label refinement or downstream generation. No large-scale manual annotation was performed.

F Task evaluation rubric

The task judge is instructed to assess a completed medical form against the supplied patient information and conversation (INPUT_SENTENCES), template (FORMAT_TEMPLATE), guidelines, and optional writing or field instructions. It is asked to assign a score between 0 and 10 for each criterion without explanation and not to confuse the criteria. The

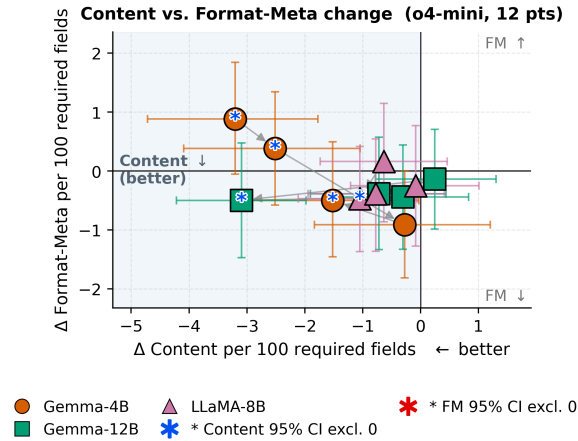


Figure 8: The o4-mini 12 points, in percentage points of the field rate, with paired 95% bootstrap CIs on both axes. The K -trajectory arrows go back and forth *without coherent direction*. The Gemma-12B trajectory is zig-zag: $\text{iter}_1 \rightarrow \text{iter}_3$ stagnation, $\text{iter}_3 \rightarrow \text{iter}_6$ lower-left spike, $\text{iter}_6 \rightarrow \text{iter}_{10}$ upper-right rebound. The horizontal CI includes $x=0$ in 7/12 points (only 5/12 are Content-*), in contrast to GPT-5.2’s 12/12 alignment (Fig. 14).

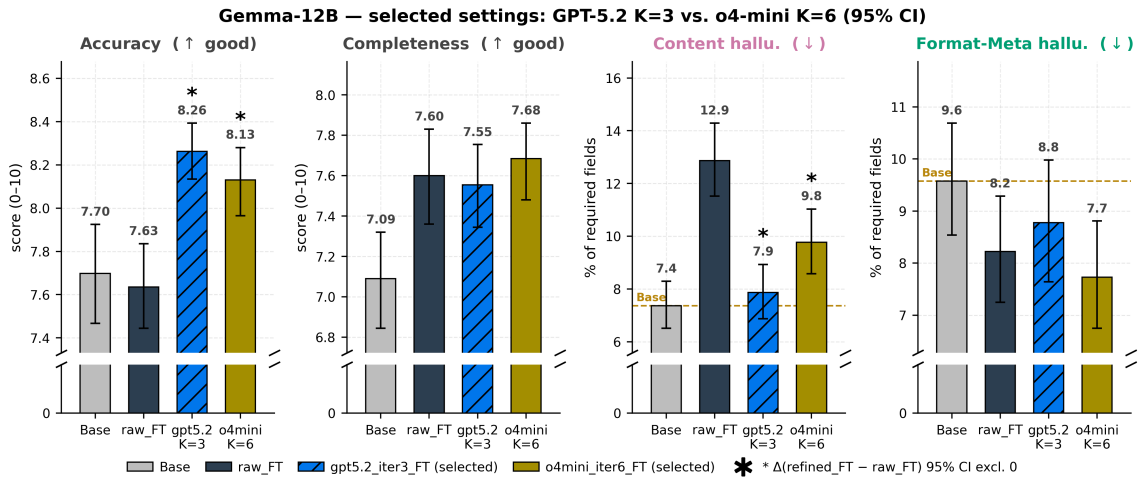


Figure 9: Absolute comparison of 4 conditions on Gemma-12B — Base / raw_FT / GPT-5.2_iter3_FT / o4mini_iter6_FT. Panels are the o3 task judge’s two rubric items and the two hallucination groups as the percentage of required fields carrying one; whiskers are 95% bootstrap CIs on each level. Both selected settings significantly reduce Content ($-5.0 / -3.1$) and significantly raise Accuracy (to 8.26/8.13) at Completeness differences whose intervals include 0. Meanwhile, FM point estimates flip sign — $+0.6$ (GPT-5.2) / -0.5 (o4-mini) — and **both 95% CIs include 0 (non-significant)**, allowing either improvement or worsening in FM.

rubric defines the endpoints below; it does not specify intermediate score anchors, per-field weights, or a fixed penalty per omission.

Accuracy. Evaluate whether the completed form accurately reflects the patient’s condition, history, and other source details without distortion or misinterpretation. Zero denotes major inaccuracies or misrepresentations; ten denotes a full and precise reflection of the original conversation.

Completeness. Evaluate whether all template fields that can be completed from the source are filled without omitting provided information. Zero

denotes that most relevant available information is missing; ten denotes that all relevant fields are completely filled whenever information is available. This criterion measures coverage of source-supported information, not the absence of fabrication.

Task-judge settings. The o3 task judge uses the default reasoning effort (medium) and no explicit maximum-output-length override. Only the supplied evaluation prompt was used: beyond the criterion descriptions and 0/10 endpoints, no intermediate-score anchors or scored examples

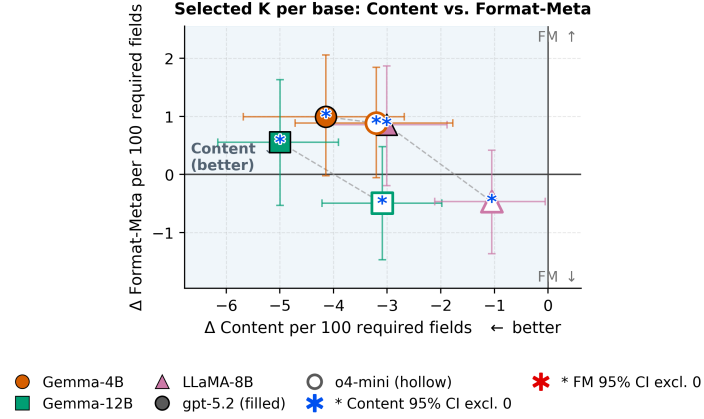


Figure 10: $(\Delta\text{Content}, \Delta\text{FM})$ scatter at per-base best- K (best = minimum point estimate of $\Delta\text{Content}$), in percentage points, with paired 95% bootstrap CIs on both axes. 6 points = 3 base $\times$ 2 refiner. Filled = GPT-5.2, hollow = o4-mini; gray dotted = the difference between the selected settings on the same base. All six selected settings have negative Content point differences and FM intervals including zero. The latter allow improvement or worsening; they do not establish unchanged formatting performance.

Base	Variant	Δ Content (pp of req. fields) ↓	Δ task
Gemma-4B	iter ₁	-3.2*	-0.04
	iter ₃	-2.5*	+0.18
	iter ₆	-0.3	+0.13
	iter ₁₀	-1.5*	-0.01
Gemma-12B	iter ₁	-0.7	+0.12
	iter ₃	+0.2	+0.33*
	iter ₆	-3.1*	+0.29*
	iter ₁₀	-0.3	+0.26*
LLaMA-8B	iter ₁	-0.6	-0.21*
	iter ₃	-1.0*	+0.03
	iter ₆	-0.1	-0.23*
	iter ₁₀	-0.8	-0.10

Table 3: 12-cell (refined_FT – raw_FT) Δ for o4-mini, in percentage points of the field rate (§4.1). Bold = lowest observed Content rate. Task differences are rounded to two decimal places from the decimal-valued source means, with ties rounded away from zero. $\Delta\text{Content}^*$ in 5/12 (vs. 12/12 for GPT-5.2); on Gemma-12B the lowest observed Content rate (iter₆) and highest o3 Task mean (iter₃) occur at different depths.

Base	Refine	Selected K	Δ Content (pp)	Δ FM (pp)
Gemma-4B	GPT-5.2	iter ₁₀	-4.1	+1.0
	o4-mini	iter ₁	-3.2	+0.9
Gemma-12B	GPT-5.2	iter ₃	-5.0	+0.6
	o4-mini	iter ₆	-3.1	-0.5
LLaMA-8B	GPT-5.2	iter ₁₀	-3.0	+0.9
	o4-mini	iter ₃	-1.0	-0.5

Table 4: Per-base settings selected by the lowest Content-rate point estimate in the evaluated grid; the same K is selected under record counts. Cells are percentages of the required fields (§4.1). All six selected $\Delta\text{Content}$ intervals exclude 0, and GPT-5.2 has a larger selected reduction than o4-mini for each base. These comparisons establish neither a general model-strength ordering nor an optimal refinement depth.

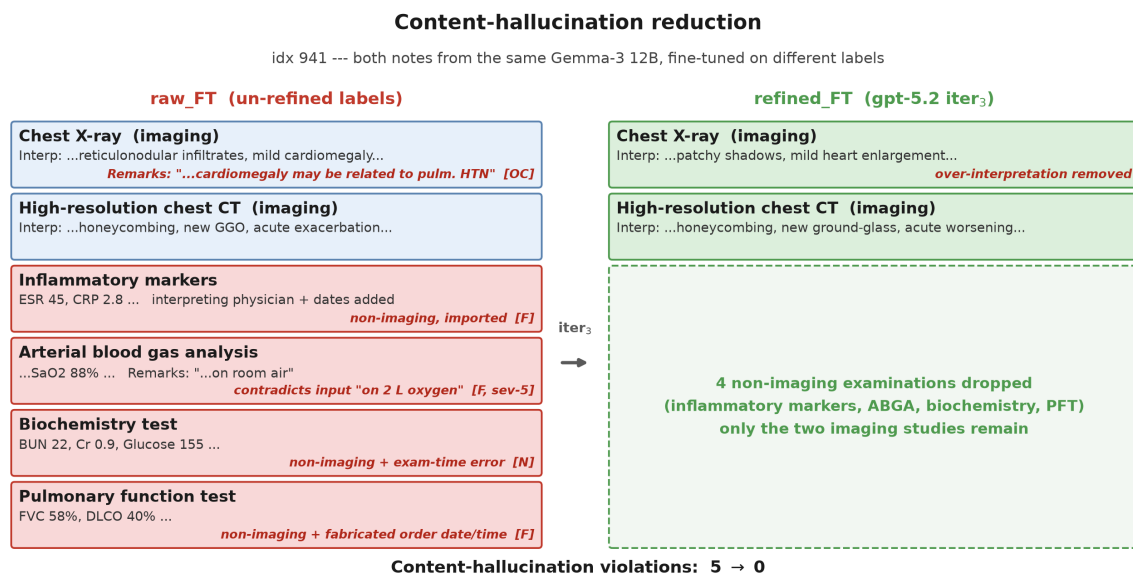


Figure 11: An illustrative case showing reduced Content hallucinations alongside increased Format-Meta hallucinations (HF idx-941). Both notes are generated by the *same* Gemma-12B, fine-tuned on different labels: raw_FT (un-refined labels) pulls four non-imaging tests (inflammatory markers, ABGA, biochemistry, PFT) into an imaging-only report and over-interprets the X-ray — 5 Content hallucinations; refined_FT (GPT-5.2 iter₃) drops the non-imaging examinations and the over-interpretation, keeping only the two imaging studies the template asks for (X-ray, HRCT), so the Content count goes 5 → 0. The o3 Completeness score remains 3/10 and FM increases from 3 to 4. Interpretation text abbreviated with “...”.

were provided. Intermediate scores were assigned by the judge; a score difference is not calibrated to a number of omitted facts or to clinical benefit.

Reported criteria. We use the Accuracy and Completeness criteria from this prompt and report them separately. These are the rubric’s endpoint descriptions, not validated clinical outcome measures.

Completeness alongside Content hallucinations. Completeness assesses coverage of available source information rather than output length. A shorter note can retain relevant information, while a longer note can repeat content without improving coverage. For Gemma-12B at iteration 3, the similar mean Completeness scores accompany substantially fewer Content hallucinations; this observation does not establish unchanged omissions. The pattern is not uniform across models or depths. Gemma-4B at iteration 1 improves Completeness by 0.290 (95% CI [0.055, 0.530]) while the Content rate decreases from 16.8% to 14.0% of the required fields. In contrast, LLaMA-8B at iteration 10 decreases Completeness by 0.290 (95% CI [−0.540, −0.030]) while the Content rate decreases from 12.7 to 9.7. Thus a lower hallucination rate can coexist with a loss in judge-assessed Completeness.

ness. Table 5 reports every GPT-5.2 setting. These exploratory, pointwise intervals are unadjusted for multiple comparisons and selection; no equivalence margin or clinically meaningful score threshold was specified.

Paired note-level analysis. We match each refined output to raw_FT by evaluation sample ID, and compute refined-minus-raw differences in each of the eight category counts, Content, Format-Meta, and Completeness. A decrease/increase is a strictly negative/positive difference; unchanged means exact equality of the stored values. No tolerance or clinical-equivalence threshold is introduced. Table 6 gives all nine joint outcomes for the selected setting; Table 7 summarizes every refiner/base/depth condition. Counts describe the fixed outputs and judge calls, not independently verified corrections of particular errors. A category count can fall even if some original errors persist and new errors appear. This analysis addresses how aggregate effects are distributed across notes, but is not a fact-level or field-level error rate, a length-controlled comparison, or an independent omission audit.

Base	K	Raw completeness	Refined completeness	Difference [95% CI]	Content, % of req. fields (raw $\rightarrow$ refined)
Gemma-4B	1	7.115	7.405	+0.290 [+0.055, +0.530]	16.8 $\rightarrow$ 14.0
Gemma-4B	3	7.115	7.233	+0.117 [-0.138, +0.373]	16.8 $\rightarrow$ 13.2
Gemma-4B	6	7.115	7.320	+0.205 [-0.050, +0.465]	16.8 $\rightarrow$ 12.8
Gemma-4B	10	7.115	7.228	+0.113 [-0.122, +0.355]	16.8 $\rightarrow$ 12.7
Gemma-12B	1	7.600	7.603	+0.003 [-0.253, +0.258]	12.9 $\rightarrow$ 11.3
Gemma-12B	3	7.600	7.555	-0.045 [-0.305, +0.230]	12.9 $\rightarrow$ 7.9
Gemma-12B	6	7.600	7.590	-0.010 [-0.240, +0.225]	12.9 $\rightarrow$ 8.1
Gemma-12B	10	7.600	7.645	+0.045 [-0.200, +0.295]	12.9 $\rightarrow$ 8.4
LLaMA-8B	1	7.525	7.568	+0.043 [-0.195, +0.287]	12.7 $\rightarrow$ 10.9
LLaMA-8B	3	7.525	7.393	-0.133 [-0.378, +0.125]	12.7 $\rightarrow$ 10.4
LLaMA-8B	6	7.525	7.380	-0.145 [-0.390, +0.100]	12.7 $\rightarrow$ 10.3
LLaMA-8B	10	7.525	7.235	-0.290 [-0.540, -0.030]	12.7 $\rightarrow$ 9.7

Table 5: Completeness and the Content rate across all 12 GPT-5.2 settings on the same 200 evaluation examples; Content is the percentage of required fields carrying a hallucination (§4.1). Completeness differences compare refined_FT with raw_FT; intervals use the additional 10,000-resample paired bootstrap described in the main text. Higher Completeness and a lower Content rate are desirable. An interval containing zero does not establish equivalence.

Content change	Completeness change		
	Lower	Equal	Higher
Fewer	43	37	34
Equal	21	21	19
More	14	5	6

Table 6: Paired changes for Gemma-12B/GPT-5.2 iteration 3 versus raw_FT ($n = 200$). Equal Completeness denotes equal judge scores, not proven information preservation.

G Training configuration

Training is implemented using Axolotl 0.14.0. Within each base model, raw-label SFT and refined-label SFT at every evaluated depth and for both refiners use the identical configuration below. Five settings are tuned per base model — adapter rank, alpha, and dropout, learning rate, and gradient accumulation — and are 64/128/0.1, 10^{-4} , and 4 for Gemma-12B; 16/32/0.1, 2×10^{-4} , and 8 for Gemma-4B; and 64/128/0.05, 5×10^{-5} , and 8 for LLaMA-8B. All remaining settings are shared. Model and tokenizer checkpoints, chat templates, dataset paths, and output paths vary with the experimental condition. Realized training duration depends on the stopping rule in each run.

Adapter and precision. The model is loaded in 4-bit NF4 with double quantization, using bfloat16 for the 4-bit compute and storage dtypes. The adapter setting is QLoRA with the per-base rank, alpha, and dropout listed above. Target modules are q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj. The precision flags are

bf16=auto, fp16=false, and tf32=true.

Input and batching. The configuration uses each model’s corresponding chat template and end-of-turn token, assistant-role training, last end-of-sequence token supervision, and train_on_inputs=false. Maximum sequence length is 8,192. Training and evaluation sample packing and padding to the maximum sequence length are disabled. Training uses one NVIDIA RTX PRO 6000 GPU. Micro-batch size is 2 for every base model, so the per-base gradient accumulation above gives an effective batch size of 8 examples per optimizer update for Gemma-12B and 16 for Gemma-4B and LLaMA-8B. Gradient checkpointing is enabled with use_reentrant=false; Flash Attention is disabled.

Optimization. The configured seed is 42. Optimization uses fused AdamW (adamw_torch_fused), the per-base learning rate above, a cosine schedule, weight decay 0.01, warmup ratio 0.03, and maximum gradient norm 1.0. The epoch limit is 7.

Evaluation-note generation. All Base, raw_FT, and refined_FT evaluation notes are generated with the same sampling settings: temperature 0.7 and top- p 0.95. Failed generations and empty responses are retried until a non-empty response is obtained; generated notes are judged as produced, without exclusion or regeneration.

Base	Refiner	K	Content change			Completeness among notes with fewer Content hallucinations		Higher
			Fewer	Equal	More	Lower	Equal	
Gemma-4B	GPT-5.2	1	90	59	51	28	32	30
Gemma-4B	GPT-5.2	3	98	61	41	33	29	36
Gemma-4B	GPT-5.2	6	101	52	47	38	26	37
Gemma-4B	GPT-5.2	10	99	53	48	35	23	41
Gemma-4B	o4-mini	1	93	54	53	36	31	26
Gemma-4B	o4-mini	3	91	47	62	26	32	33
Gemma-4B	o4-mini	6	71	52	77	23	30	18
Gemma-4B	o4-mini	10	85	48	67	33	22	30
Gemma-12B	GPT-5.2	1	90	58	52	36	25	29
Gemma-12B	GPT-5.2	3	114	61	25	43	37	34
Gemma-12B	GPT-5.2	6	118	49	33	43	39	36
Gemma-12B	GPT-5.2	10	109	56	35	42	34	33
Gemma-12B	o4-mini	1	86	50	64	29	27	30
Gemma-12B	o4-mini	3	72	57	71	26	28	18
Gemma-12B	o4-mini	6	105	52	43	42	32	31
Gemma-12B	o4-mini	10	68	60	72	19	21	28
LLaMA-8B	GPT-5.2	1	99	50	51	37	33	29
LLaMA-8B	GPT-5.2	3	96	52	52	47	28	21
LLaMA-8B	GPT-5.2	6	95	56	49	35	34	26
LLaMA-8B	GPT-5.2	10	103	50	47	48	27	28
LLaMA-8B	o4-mini	1	82	50	68	39	25	18
LLaMA-8B	o4-mini	3	80	64	56	33	25	22
LLaMA-8B	o4-mini	6	70	59	71	26	26	18
LLaMA-8B	o4-mini	10	81	49	70	38	22	21

Table 7: Paired note-level changes versus raw_FT across all 24 refinement conditions (200 notes each). Fewer, equal, and more Content counts sum to 200 per row. The final three columns partition only the notes with fewer Content hallucinations and sum to the Fewer column. These are descriptive counts without multiplicity-adjusted inference.

Checkpoint selection. Evaluation and checkpoint saving are configured every 20 steps, with at most five checkpoints retained. The model with the lowest eval_loss is loaded at the end. Early stopping has patience 3 and threshold 0.001.

Internal validation setting. With val_set_size=0.1, the 500-example dataset snapshot is partitioned into 450 optimization and 50 internal-validation examples. The internal set is used for early stopping and minimum-loss checkpoint selection. The 450/50 sample split is fixed across all three base models, raw-label SFT, both refiners, and every evaluated depth; only the completed-note labels change with refinement. The 200-example evaluation set is randomly drawn without generator stratification from the same original 3,000-example pool and is disjoint from the entire 500-example refinement corpus. The random seed for selecting the refinement and evaluation subsets was 123, distinct from the training seed of 42.

H Detection and refinement prompts

Shared task specification. The hospital-form completion prompt is inserted into both detection and refinement. It supplies three components: user instructions, a medical template, and patient information. Instructions are hard constraints, patient information is the only factual source, and the template fixes field names, headings, order, line breaks, tables, and separators. Placeholder or example content must be replaced with source-supported content. When required information is missing, the prompt follows the user’s missing-data instructions; absent such instructions, it permits specified placeholders such as a dash, “No record,” “No notable findings,” or “Not applicable.” It prohibits unsupported inference and requests only the completed template. Adherence to these instructions is assessed through the output evaluations.

Detection inputs and outputs. The system message supplies the error-classification instructions, the eight category definitions and examples, and the shared task prompt. The user message supplies the original task input and current completed-note output. For each hallucination, the detector is asked for

its type and a verbatim sentence or field with an explanation. The call requires the VIOLATION_TOOLS interface with `tool_choice=required`.

Structured output and counting unit. The function VIOLATION_SCHEMA returns an object containing a violations array. Each item carries a single violation_type chosen from the eight categories together with a free-text violated_reason; additional properties are disallowed at both object levels. There is no separate evidence field, although the prompt requests a verbatim excerpt and explanation. The schema permits an empty array and does not enforce uniqueness across records. Multiple records can therefore refer to the same sentence or fact; single-category assignment per record does not establish mutually exclusive underlying error categories. All returned hallucination records are counted without deduplication. Counts therefore measure detector-returned records rather than unique clinical errors. The schema field names use *violation*; each such record is a hallucination record in the terminology of §4.1.

Training-label evaluation. For both refiners, the reported training-label hallucination counts at every round, including round 0, come from Gemini evaluation of the labels. Gemini results are stored for reporting only and are not fed back into refinement or early stopping; both use the refiner's own DETECT records. A common judge and rubric support comparisons across rounds; rubric validity is discussed in Limitations.

Condition information. Neither the hallucination judge nor the task judge is given the generating model name, the Base/raw_FT/refined_FT condition, or the refinement iteration. Withholding these identifiers does not rule out sensitivity to output style or shared-rubric bias.

Evaluation repetitions. Each note receives one valid response from each judge; scores and hallucination records are not aggregated across repeated judgments. Failure retries are distinct from repeated valid evaluations. The reported bootstrap intervals condition on these fixed responses and do not quantify judge-call variability.

Call failures and retries. For both the hallucination and task judges, failed calls and responses that violate the required format are retried until a valid response is obtained. DETECT and REFINE use the same retry policy for call failures and invalid

response formats. A valid empty violations array is a zero-detection result, distinct from a failed or malformed response.

Shared detection protocol. The same detection prompt and tool interface are used by the self-detection node and the external Gemini evaluation node. For self-detection, GPT-5.2 uses high reasoning effort and o4-mini uses medium reasoning effort (the default setting used in these runs); both have a configured output-token limit of 16,384. The evaluation model is gemini-3.1-pro-preview with high thinking level and an output-token limit of 8,192. The judge uses a different model with the same classification instructions and task specification.

Category interpretation. Factual errors concern invented facts; Medical Reasoning concerns unsupported clinical causality or judgments; Conversational Context concerns dialogue interpretation and speaker attribution; Numerical concerns numbers, times, ranges, or units. Structural includes incorrect field placement, invented fields, and duplicated or deleted required fields. Metadata / Environment concerns internal prompts, system information, or the model's generation context, such as identifying the document as GPT-generated. Prompt Overgeneralization concerns copying or generalizing prompt examples without patient-specific support. Overclinical Reasoning concerns inference using medical knowledge absent from the dialogue. Category assignment used only these definitions, with no additional tie-breaking rules for Medical Reasoning versus Overclinical Reasoning. For example, the rubric's unsupported "suspected pneumonia" inference from cough and mild fever could fit both definitions; this illustrates conceptual overlap, not an adjudicated example from the evaluation set. These definitions can overlap, and they do not specify a clinically validated ordering of severity.

Refinement inputs and constraints. The refiner receives the shared task prompt, original task input, current note, and detected error list. It is instructed to preserve all correct content, revise or remove only listed issues, make minimal corrections, preserve style and structure as much as possible, and return only the corrected note. The task specification, detection interface, category definitions, and refinement constraints are described in this appendix. For REFINE calls, GPT-5.2 uses high reasoning effort and o4-mini uses medium reason-

ing effort (the default setting used in these runs); both have a maximum output of 16,384 tokens.

I Output-length sensitivity analysis

This appendix asks whether the hallucination-rate reductions of §4.3 are produced by shorter, more conservative completions, and reports output lengths and length-normalized counts for every condition. Lengths are character counts of the generated notes, 200 per condition for all 30 conditions. Table 8 reports mean and median lengths alongside Content counts.

We estimate corpus-level Content hallucination density as $1,000C/(n\bar{L})$, where C is the total Content hallucination count, $n = 200$, and $\bar{L}$ is mean character length. This ratio of totals is distinct from a mean of note-level densities. Per-sample lengths are available for all 30 conditions, so we accompany each refined-minus-raw density difference with a paired percentile bootstrap interval that resamples the same 200 evaluation IDs across the compared conditions (10,000 resamples; seed 20260907), recalculating the ratio in each resample. Intervals are pointwise and conditional on the fixed outputs and judge records, without correction for multiple comparisons. This normalization is a descriptive exposure adjustment, not a matched-length experiment.

GPT-5.2 refinement lowers density relative to raw_FT in all 12 settings (5.8–44.6%), and the interval excludes zero in 7 of the 12. For Gemma-12B at iteration 3, mean length falls from 2,310.8 to 1,979.7 characters while Content hallucination density falls from 1.086 to 0.768, a difference of -0.318 (95% CI $[-0.443, -0.199]$). In Gemma-4B, mean lengths increase by 7.4–37.6%, Content counts decrease by 17.0–24.9%, both task judges’ mean Completeness increases at all four depths, and the density reduction is significant at every depth; between 55.5% and 68.5% of individual outputs are longer than their raw_FT counterparts. Output shortening alone therefore does not explain these reductions. With o4-mini, density decreases in six settings and increases in six; no decrease has an interval excluding zero, whereas four increases do.

Length distributions are skewed. LLaMA-8B/o4-mini iteration 1 has a mean of 4,118.4 characters but a median of 1,724.5, and only 31.5% of its notes are longer than their raw_FT counterparts. Gemma-12B/o4-mini iteration 3 likewise has a higher mean

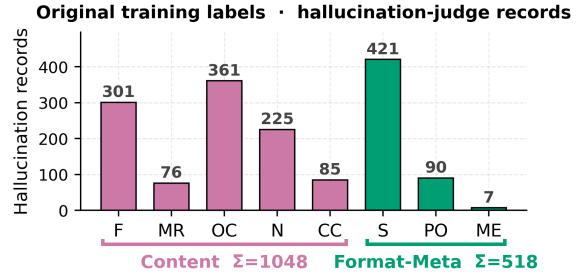


Figure 12: Hallucination-judge 8-cat distribution of synthetic training labels ($n=500$, $\Sigma=1,566$, ≈ 3.1 per sample). Bar color = 2-group taxonomy (pink = Content / green = Format-Meta); brackets below the x -axis = group sums.

but a lower median than raw_FT. A few long outputs can lower corpus-level density without improving factual quality. Means and medians are reported without trimming. Character normalization does not measure supported-fact recall or field coverage; task-judge Completeness and an independent omission audit would address different questions.

J Task-component comparison under both judges

Table 9 expands the raw_FT — Base task-component comparison of §4.2.

K Supplementary training and cost plots

The following plots and table provide supporting detail for the training-label analysis, joint Content/Format-Meta changes, and refinement costs discussed in the main text.

Cost calculation. For a stored response, let I , C , and O denote API-reported input, cached-input, and output tokens. The calculation is $[(I - C)p_{\text{in}} + Cp_{\text{cache}} + Op_{\text{out}}]/10^6$. The rates used (USD per million tokens, in that order) are (1.75, 0.175, 14.00) for GPT-5.2 and (1.10, 0.275, 4.40) for o4-mini. Usage from stored DETECT and REFINE responses is summed; evaluation responses are excluded. Each refinement iteration pairs a detection with the refinement it triggers, and iteration 0 has no refinement-call cost. The reported totals sum every successful REFINE and DETECT call made during the refinement run: for depth K these are $\text{REFINE}_1\text{--REFINE}_K$ together with the detections $\text{DETECT}_1\text{--DETECT}_K$ that evaluate their outputs and determine whether refinement continues. The detection of the original note (DETECT_0) was produced before these runs

Base	Condition	Mean chars	Median chars	Content	Content/1k chars	Δ density (%)
Gemma-4B	Base	2,247.7	1,721.0	366	0.814	—
Gemma-4B	raw_FT	2,333.5	2,075.5	658	1.410	—
Gemma-4B	GPT-5.2, $K = 1$	2,634.5	2,194.0	546	1.036	-26.50
Gemma-4B	GPT-5.2, $K = 3$	2,687.4	2,207.0	519	0.966	-31.51
Gemma-4B	GPT-5.2, $K = 6$	3,210.8	2,326.5	502	0.782	-44.55
Gemma-4B	GPT-5.2, $K = 10$	2,506.8	2,184.5	494	0.985	-30.11
Gemma-4B	o4-mini, $K = 1$	2,082.1	1,728.0	532	1.278	-9.39
Gemma-4B	o4-mini, $K = 3$	2,141.1	1,915.0	551	1.287	-8.74
Gemma-4B	o4-mini, $K = 6$	1,947.4	1,906.0	637	1.636	+16.00
Gemma-4B	o4-mini, $K = 10$	2,285.3	1,873.5	595	1.302	-7.67
Gemma-12B	Base	1,726.4	1,424.5	277	0.802	—
Gemma-12B	raw_FT	2,310.8	2,029.5	502	1.086	—
Gemma-12B	GPT-5.2, $K = 1$	2,244.9	2,052.5	439	0.978	-9.98
Gemma-12B	GPT-5.2, $K = 3$	1,979.7	1,807.5	304	0.768	-29.31
Gemma-12B	GPT-5.2, $K = 6$	2,040.4	1,837.5	308	0.755	-30.51
Gemma-12B	GPT-5.2, $K = 10$	1,946.2	1,813.5	317	0.814	-25.02
Gemma-12B	o4-mini, $K = 1$	2,115.7	1,968.0	460	1.087	+0.08
Gemma-12B	o4-mini, $K = 3$	2,495.8	1,972.5	500	1.002	-7.78
Gemma-12B	o4-mini, $K = 6$	1,848.0	1,766.5	374	1.012	-6.84
Gemma-12B	o4-mini, $K = 10$	1,997.2	1,934.0	485	1.214	+11.78
LLaMA-8B	Base	2,256.1	1,819.0	385	0.853	—
LLaMA-8B	raw_FT	2,369.9	1,970.5	486	1.025	—
LLaMA-8B	GPT-5.2, $K = 1$	2,158.4	1,785.5	412	0.954	-6.92
LLaMA-8B	GPT-5.2, $K = 3$	2,152.0	1,831.0	402	0.934	-8.91
LLaMA-8B	GPT-5.2, $K = 6$	2,003.3	1,834.5	387	0.966	-5.80
LLaMA-8B	GPT-5.2, $K = 10$	2,282.2	1,882.5	374	0.819	-20.09
LLaMA-8B	o4-mini, $K = 1$	4,118.4	1,724.5	470	0.571	-44.35
LLaMA-8B	o4-mini, $K = 3$	2,089.6	1,829.0	437	1.046	+1.98
LLaMA-8B	o4-mini, $K = 6$	1,788.2	1,625.0	474	1.325	+29.26
LLaMA-8B	o4-mini, $K = 10$	1,743.8	1,667.5	457	1.310	+27.79

Table 8: Character-length aggregates of the generated notes and Content density for all 30 conditions (200 outputs each). Density changes compare refined_FT with raw_FT of the same base model; paired bootstrap intervals for the twelve GPT-5.2 differences are reported in the text.

Base model	Component	o3	Gemini 3.8 Flash
Gemma-4B	Accuracy	+0.230 [-0.097, +0.503]	+0.572* [+0.242, +0.868]
	Completeness	+1.250* [+0.960, +1.535]	+1.199* [+0.869, +1.520]
Gemma-12B	Accuracy	-0.062 [-0.315, +0.218]	-0.181 [-0.463, +0.112]
	Completeness	+0.510* [+0.245, +0.775]	+0.179 [-0.113, +0.459]
LLaMA-8B	Accuracy	+0.742* [+0.525, +0.988]	+0.944* [+0.681, +1.228]
	Completeness	+1.130* [+0.885, +1.375]	+1.006* [+0.775, +1.237]

Table 9: Accuracy and Completeness for raw_FT — Base under both task judges on the same 200 evaluation samples. Cell = Δ [low, high]; * = the interval excludes 0. The o3 intervals are those plotted in Fig. 2; the Gemini intervals apply the same paired procedure to that judge’s stored per-note scores ($n_{\text{boot}}=1,000$, seed 0, percentile [2.5%, 97.5%]). Completeness increases on all 3 bases under o3, whereas Accuracy does so on 1 of 3; the two judges agree in sign on all six component differences. An interval containing zero does not establish equivalence.

and is not part of the reported totals. A terminating DETECT that returns no hallucinations is therefore counted even though no REFINE follows it. Usage fields were present in every included response; the rates are those used in our cost calculation, not a reconciliation with billing statements.

L Notes with no detected hallucinations

We report the number and proportion of notes with zero detected Content hallucination records and zero records across all eight categories (Table 11). Each condition contains the same 200 evaluation inputs. For the selected Gemma-12B iter₃ setting, the Content-zero count rises from 29 of 200 (14.5%) under raw_FT to 54 (27.0%), against 51 (25.5%) for Base; across the 12 GPT-5.2 settings

Hallucination-judge records in training labels (iterations 0-10, n=500 labels)

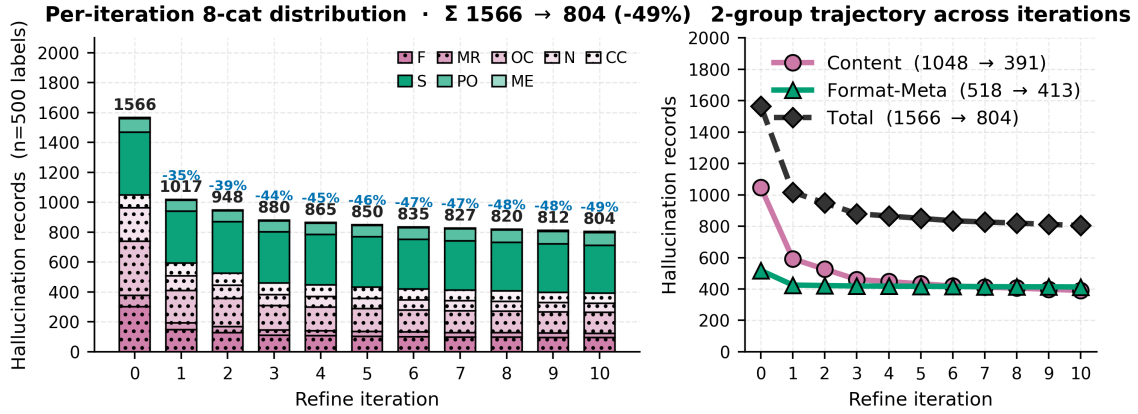


Figure 13: Hallucination-judge 8-cat evolution of the training labels ($n=500$) over GPT-5.2 refinement iterations $0, \dots, 10$. **(Left)** 8-cat stacked bar (2-group palette: Content pink (*dotted*) / Format-Meta green + within-group shade); bar-top label = $\Delta\%$ vs iteration 0. **(Right)** 2-group trajectory: Content (pink ●), FM (green ▲), Total (black ◇ dashed).

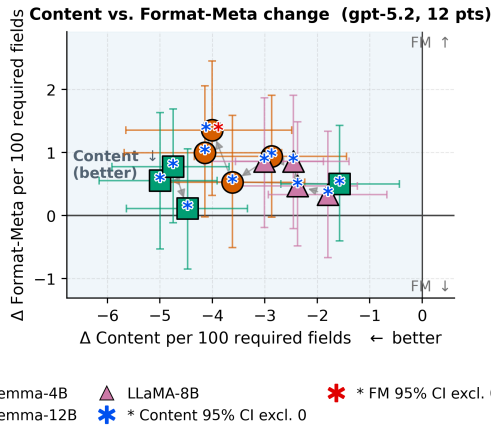


Figure 14: $x=\Delta$ Content, $y=\Delta$ FM (both refined_FT – raw_FT, in percentage points), with paired 95% bootstrap CI bars on both axes. 12 points = $3 \text{ base} \times \text{GPT-5.2} \times K \in \{1, 3, 6, 10\}$. **All 12 points have $x < 0$ with horizontal CI excluding 0 (Content significantly reduced for every cell)**; FM increases significantly for Gemma-4B at $K = 6$; the other eleven vertical CIs include zero and do not establish equivalence. Arrows = K trajectory.

Training-label hallucination records and recorded costs DETECT + REFINE ($n=500$)

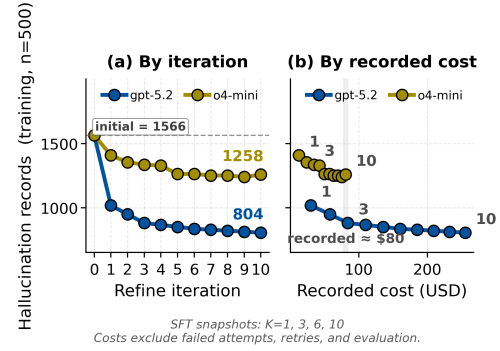


Figure 15: **(a)** Per-iteration *train-side* residual hallucinations remaining in the labels — GPT-5.2 drops steeply from iteration $0 \rightarrow 1$ (1,566 $\rightarrow$ 1,017); o4-mini iteration 3 = 1,336 and iteration 10 = 1,258 plateau. **(b)** x = cumulative refinement cost, y = train-side residual hallucinations (sharey). At approximately \$80 in recorded successful-response costs (failed attempts, retries, and evaluation costs excluded), **GPT-5.2 iteration 3 (880) leaves fewer train residuals than o4-mini iteration 10 (1,258)**.

M Additional task-judge sensitivity analysis

it increases relative to raw_FT in 10 (Gemma-4B iter₃ and LLaMA-8B iter₁ decrease). All-category zero proportions remain between 0% and 2% across all 30 settings. These descriptive proportions complement the field-normalized rates but do not establish clinical error freedom or judge sensitivity.

Gemini 3.8 Flash re-evaluates the same generated notes for the same 200 examples with the same evaluation prompt as o3, and we again use its Accuracy and Completeness criteria. The additional evaluation covers all 30 conditions: Base, raw_FT, and four refinement depths for each of GPT-5.2 and o4-mini, across three base models. It uses the

Refine	Cumulative cost (USD)	Residual hallu
GPT-5.2 (iter 1–3)	\$83.83	880
o4-mini (iter 1–3)	\$34.61	1,336
o4-mini (iter 1–10)	\$80.56	1,258

Table 10: Cumulative costs of final successful DETECT and REFINE responses and *train-side* residual hallucination records in 500 labels. Costs are calculated from API-reported token usage and model-specific rates. Failed attempts, retries, and evaluation are excluded; these are not full execution costs. At approximately \$80 in recorded costs, GPT-5.2 iteration 3 leaves fewer residual records than o4-mini iteration 10 (880 vs. 1,258). The three rows compare settings at comparable recorded cost rather than at equal depth: GPT-5.2 iteration 10 reaches \$255.00 with 804 residual records. This is a comparison of label-refinement outcomes and API costs, not a measure of downstream or clinical cost-effectiveness. Figure 15 shows the observed cost trajectories; Appendix C reports the selected per-base comparisons.

default thinking level (medium) and no explicit maximum-output-length override. One valid response per note supplies the final score.

Relative to raw_FT, Accuracy means increase under both judges in 22/24 refined conditions. Completeness increases under both in 13/24, changes in opposite directions in 6/24, and decreases under both in 5/24. These are descriptive comparisons of means.

For GPT-5.2 refinement, all four Gemma-4B conditions improve Completeness under both judges, whereas LLaMA-8B at $K = 10$ declines under both. For Gemma-12B at $K = 3$, the o3 difference is -0.045 (95% CI $[-0.305, 0.230]$), whereas Gemini’s mean difference is $+0.508$. For o4-mini refinement, Gemma-12B improves Accuracy and Completeness under both judges at all four depths. At $K = 6$, its Content count falls from 502 to 374, with Completeness differences of $+0.085$ (o3) and $+0.405$ (Gemini). However, o4-mini’s Gemma-4B at $K = 10$ and LLaMA-8B at $K = 1, 6, 10$ show lower Completeness under both judges. Thus a uniform loss of judge-assessed Completeness does not explain all observed Content reductions, but omissions and length effects remain unresolved.

Table 12 reports all condition means. Changes of means do not establish equivalence or clinical benefit, and the intervals below are pointwise and unadjusted for the 24 comparisons. Shared prompts and one call per note also leave rubric bias and

repeated-judgment variability unresolved.

Paired intervals under the additional judge.

This judge returns one score per note, so the paired bootstrap of §4.1 applies to it unchanged ($n_{\text{boot}}=10,000$, seed 20260907, applied to both judges’ stored per-note scores for this comparison). Over the 24 refined conditions, Accuracy increases $\ast$ in 17 and Completeness in 10, and *no* condition shows a $\ast$ Completeness decrease; the corresponding o3 counts are 12 $\ast$ Accuracy increases and 3 $\ast$ Completeness increases against 2 $\ast$ Completeness decreases. For the selected Gemma-12B $\times$ GPT-5.2 iter₃ configuration both components rise $\ast$ under this judge — Accuracy $+0.947^\ast$ [$+0.705, +1.209$] and Completeness $+0.508^\ast$ [$+0.247, +0.792$] — whereas o3 leaves Completeness inconclusive (-0.045 [$-0.305, +0.230$]). Restricted to GPT-5.2, Accuracy is $\ast$ in 11 of 12 cells (all but LLaMA-8B iter₁₀) and Completeness in 5 of 12, against 8 and 1 under o3. The Accuracy gain is thus reproduced under both judges, while the coverage conclusion is judge-dependent: one judge supports an increase at the selected setting and neither supports a decrease. Several Completeness intervals lie close to zero, so these counts shift by one or two cells under a smaller resample count or a different seed.

Level shift and note-level agreement. The additional judge’s higher absolute scores are concentrated in Completeness: averaged over all 30 conditions and 200 notes, its scores exceed o3’s by $+1.145$ on Completeness against $+0.475$ on Accuracy, so the gap is not a uniform rescaling of both components. Per-note scores correlate at $r=0.761$ (Accuracy) and $r=0.685$ (Completeness) across the 6,000 scored notes and land within one point of each other on 76.6% and 57.9% of them, yet the two judges agree on the *sign* of the per-note change against raw_FT in only 55.2% (Accuracy) and 50.3% (Completeness) of the 4,800 paired notes. Aggregate directions therefore agree far more often than individual notes do, and note-level judgments from either judge should not be read as reliable on their own.

Base model	Condition / K	Content zero, n (%)	All-category zero, n (%)
Gemma-4B	Base	37 (18.5%)	0 (0.0%)
Gemma-4B	raw_FT	24 (12.0%)	0 (0.0%)
Gemma-4B	GPT-5.2 / 1	28 (14.0%)	0 (0.0%)
Gemma-4B	GPT-5.2 / 3	23 (11.5%)	0 (0.0%)
Gemma-4B	GPT-5.2 / 6	31 (15.5%)	0 (0.0%)
Gemma-4B	GPT-5.2 / 10	27 (13.5%)	0 (0.0%)
Gemma-4B	o4-mini / 1	27 (13.5%)	0 (0.0%)
Gemma-4B	o4-mini / 3	12 (6.0%)	0 (0.0%)
Gemma-4B	o4-mini / 6	17 (8.5%)	0 (0.0%)
Gemma-4B	o4-mini / 10	21 (10.5%)	0 (0.0%)
Gemma-12B	Base	51 (25.5%)	1 (0.5%)
Gemma-12B	raw_FT	29 (14.5%)	0 (0.0%)
Gemma-12B	GPT-5.2 / 1	30 (15.0%)	1 (0.5%)
Gemma-12B	GPT-5.2 / 3	54 (27.0%)	3 (1.5%)
Gemma-12B	GPT-5.2 / 6	55 (27.5%)	3 (1.5%)
Gemma-12B	GPT-5.2 / 10	49 (24.5%)	4 (2.0%)
Gemma-12B	o4-mini / 1	33 (16.5%)	0 (0.0%)
Gemma-12B	o4-mini / 3	27 (13.5%)	0 (0.0%)
Gemma-12B	o4-mini / 6	46 (23.0%)	2 (1.0%)
Gemma-12B	o4-mini / 10	31 (15.5%)	1 (0.5%)
LLaMA-8B	Base	27 (13.5%)	0 (0.0%)
LLaMA-8B	raw_FT	25 (12.5%)	1 (0.5%)
LLaMA-8B	GPT-5.2 / 1	22 (11.0%)	0 (0.0%)
LLaMA-8B	GPT-5.2 / 3	33 (16.5%)	0 (0.0%)
LLaMA-8B	GPT-5.2 / 6	35 (17.5%)	0 (0.0%)
LLaMA-8B	GPT-5.2 / 10	43 (21.5%)	0 (0.0%)
LLaMA-8B	o4-mini / 1	18 (9.0%)	0 (0.0%)
LLaMA-8B	o4-mini / 3	25 (12.5%)	2 (1.0%)
LLaMA-8B	o4-mini / 6	24 (12.0%)	1 (0.5%)
LLaMA-8B	o4-mini / 10	26 (13.0%)	0 (0.0%)

Table 11: Notes with no detected hallucinations, out of 200 per condition. Zero refers to the fixed judge’s returned records, not clinician-verified absence of errors.

Base model	Condition	Accuracy		Completeness	
		o3	Gemini	o3	Gemini
Gemma-4B	Base	6.733	6.694	5.865	6.808
	raw_FT	6.963	7.265	7.115	8.007
	GPT-5.2 / 1	7.488	7.908	7.405	8.478
	GPT-5.2 / 3	7.318	7.608	7.233	8.253
	GPT-5.2 / 6	7.420	7.686	7.320	8.208
	GPT-5.2 / 10	7.558	7.808	7.228	8.352
	o4-mini / 1	7.053	7.382	6.945	8.119
	o4-mini / 3	7.243	7.675	7.193	8.365
	o4-mini / 6	7.075	7.495	7.265	8.255
	o4-mini / 10	7.075	7.470	6.978	7.962
Gemma-12B	Base	7.698	8.306	7.090	8.448
	raw_FT	7.635	8.125	7.600	8.628
	GPT-5.2 / 1	8.103	8.642	7.603	8.883
	GPT-5.2 / 3	8.263	9.071	7.555	9.136
	GPT-5.2 / 6	8.268	8.913	7.590	9.030
	GPT-5.2 / 10	8.260	8.986	7.645	9.049
	o4-mini / 1	7.825	8.374	7.650	8.858
	o4-mini / 3	7.965	8.558	7.925	9.021
	o4-mini / 6	8.130	8.828	7.685	9.033
	o4-mini / 10	7.840	8.500	7.918	9.035
LLaMA-8B	Base	6.888	7.095	6.395	7.504
	raw_FT	7.630	8.039	7.525	8.510
	GPT-5.2 / 1	7.705	8.317	7.568	8.700
	GPT-5.2 / 3	7.750	8.317	7.393	8.566
	GPT-5.2 / 6	7.813	8.369	7.380	8.538
	GPT-5.2 / 10	7.728	8.139	7.235	8.300
	o4-mini / 1	7.525	8.045	7.215	8.375
	o4-mini / 3	7.823	8.301	7.400	8.545
	o4-mini / 6	7.428	8.043	7.270	8.409
	o4-mini / 10	7.660	8.139	7.305	8.483

Table 12: Accuracy and Completeness under o3 and Gemini 3.8 Flash on the same 200 outputs per condition, covering both refiners. All scores are higher-is-better, on a 0–10 scale. Means are rounded to three decimals. These are descriptive means, not significance comparisons.